SYMMETRY OF MONOTONE TRAVELING WAVES FOR THE FRACTIONAL ALLEN–CAHN EQUATION IN THE PLANE WITH $1/2 < s < 1$

Abstract. We prove horizontal reflection symmetry and strict horizontal monotonicity for vertically increasing traveling waves of the fractional Allen–Cahn equation in the plane, with $1/2 < s < 1$ and a balanced nondegenerate double-well potential. Only the two pointwise vertical end states are prescribed. The proof first derives finite dissipation and the upward two-arm geometry. A linearized Liouville theorem then determines the interaction of the two layers and the power law for their separation. Weighted error estimates and an extension stress identity show that the midpoint converges to a finite limit. This limit determines the reflection axis in a global moving-plane argument.

Draft status. This candidate proof was obtained by our team on September 07, 2026, through extensive dialogue-based training with AI. The team is currently working hard to verify all the proofs.

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1. Introduction

We consider bounded traveling profiles for the fractional Allen–Cahn equation in the plane. The propagation direction is denoted by y, and x is the transverse coordinate. We assume $1/2 < s < 1$, $c > 0$, and $F \in C^{3,\beta}_{\mathrm{loc}}(\mathbb{R})$ for some $0 < \beta < 1$, with

$$\begin{gathered} F(-1) = F(1) = F'(-1) = F'(1) = 0, \qquad F''(\pm 1) > 0, \\ F(t) > 0 \quad (-1 < t < 1), \\ F'(t) > 0 \quad (-1 < t < 0), \qquad F'(t) < 0 \quad (0 < t < 1). \end{gathered} \tag{1.1}$$

The traveling-wave equation is

$$(-\Delta)^s u - c u_y + F'(u) = 0 \qquad \text{in } \mathbb{R}^2, \tag{1.2}$$

where the fractional Laplacian has Fourier symbol $|\xi|^{2s}$. The vertical end states are prescribed pointwise:

$$\lim_{y \to -\infty} u(x,y) = -1, \qquad \lim_{y \to +\infty} u(x,y) = 1 \quad \text{for every fixed } x \in \mathbb{R}. \tag{1.3}$$

For the local Allen–Cahn equation, Gui proved reflection symmetry and strict transverse monotonicity for vertically monotone traveling waves with these end states [4, Theorem 1.1]. In the

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fractional equation, the algebraic tails of the layers couple the two distant arms. Determining their separation does not by itself locate a fixed symmetry axis: one must also control the accumulated motion of their midpoint. Our main result is the following.

Theorem 1.1. *Let $1/2 < s < 1$, $c > 0$, and let $F \in C^{3,\beta}_{\mathrm{loc}}(\mathbb{R})$, $0 < \beta < 1$, satisfy (1.1). Suppose $u \in C^2(\mathbb{R}^2) \cap L^\infty(\mathbb{R}^2)$ is a classical solution of (1.2), with*

$$|u| < 1, \qquad u_y > 0 \quad \text{in } \mathbb{R}^2,$$

and with the pointwise limits (1.3). Then there is $b \in \mathbb{R}$ such that

$$u(2b - x, y) = u(x, y) \quad \text{for all } (x, y) \in \mathbb{R}^2, \qquad u_x(x, y) < 0 \quad \text{for } x > b.$$

The geometry at infinity and its quantitative scale are consequences of the same hypotheses. To state them, let g be the increasing one-dimensional layer normalized by

$$(-\partial_{xx})^s g + F'(g) = 0, \qquad g(0) = 0, \qquad g(\pm\infty) = \pm 1,$$

and set

$$M = \int_{\mathbb{R}} (g')^2 \, dx, \qquad C_{1,s} = \frac{\Gamma(1 + 2s) \sin(\pi s)}{\pi}, \qquad \nu_s = \frac{2C_{1,s}}{scM}. \tag{1.4}$$

The assumptions allow different curvatures at the two wells.

Theorem 1.2. *Under the hypotheses of Theorem 1.1, there is Y such that, for $y > Y$, the zero set has two branches $\ell_1^0(y) < \ell_2^0(y)$. There are C^2 centers $\ell_1(y) < \ell_2(y)$ for which, with*

$$\Phi(x, y) = g(x - \ell_1(y)) + g(\ell_2(y) - x) - 1, \qquad v = u - \Phi, \qquad Z_1 = g'(x - \ell_1), \quad Z_2 = g'(\ell_2 - x),$$

one has $\int_{\mathbb{R}} v Z_i \, dx = 0$ and

$$\ell_i - \ell_i^0 \longrightarrow 0, \qquad \ell_1 \longrightarrow -\infty, \qquad \ell_2 \longrightarrow +\infty.$$

Writing $d = \ell_2 - \ell_1$ and $m = (\ell_1 + \ell_2)/2$, one has, as $y \to \infty$,

$$\|u(\cdot, y) - \Phi(\cdot, y)\|_\infty \le C d(y)^{-2s}, \tag{1.5}$$

$$\ell_1' = -\nu_s d^{-2s} + o(d^{-2s}), \qquad \ell_2' = \nu_s d^{-2s} + o(d^{-2s}), \tag{1.6}$$

$$|\ell_1''| + |\ell_2''| \le C d^{-2s}, \tag{1.7}$$

$$d(y)^{1+2s} = \frac{4(1 + 2s)C_{1,s}}{scM}\, y + o(y), \tag{1.8}$$

$$m(y) \longrightarrow b, \tag{1.9}$$

where b is the symmetry axis in Theorem 1.1.

Outline of the proof. The proof begins with the weighted harmonic extension of u. Its mixed stress has a bounded transverse flux, whose derivative is $-d_s c \int u_y^2 \, dy$. This gives finite total dissipation and makes horizontal translates stationary at infinity. A separate half-plane stress identity transfers finite energy to these stationary limits. They can then be classified as the one-dimensional layers or the constant wells. Thus every remote transition is locally a layer, and every nonempty horizontal slice of a far transition band is an interval of controlled width. Stable-phase barriers turn this geometric information into uniform slice bounds and extension-energy compactness. The slice Hamiltonian is strictly increasing in y; its endpoint values select the upward two-arm configuration. These arguments occupy Sections 2–3.

At this point, the profile is uniformly approximated by two layers, but the approximation and the interface speeds are only $o(1)$. Orthogonality to the two translated copies of g' fixes the centers. Section 4 develops the linear estimate that improves this qualitative description. An entire solution of the linearized equation with at most linear growth in y is affine in y along the translation mode. To prove this, a positive massive kernel first places every nonzero temporal frequency in L^2 in x; the spectral gap then eliminates that frequency. The remaining affine coefficient is determined by a scalar compatibility identity.

Section 5 applies this classification to the error after subtracting the complete frozen pair. Subtracting both layers retains the distant layer's contribution at the scale d^{-2s}. A long-interval

selection first rules out errors larger than this scale. At the interaction scale, the exact layer tails yield the forcing, and its projection on g' gives the two signed velocities in (1.6). Their difference gives (1.8).

The leading velocities cancel in m', but their first-order asymptotics alone do not imply convergence of m. This is the remaining obstruction to reflection. In Section 6, an explicit correction of the exterior tails gives weighted spatial bounds for the error and its derivatives, also when the two well curvatures differ. Orthogonality then cancels the linear terms in the boundary mixed product. With U denoting the extension and $a = 1 - 2s$, the relevant identities are

$$Q(y) := \int_{\mathbb{R}} u_x u_y \, dx = -2Mm'(y) + O(d^{-4s}), \qquad \Pi(y) := \int_0^\infty \int_{\mathbb{R}} z^a U_x U_y \, dx \, dz, \qquad \Pi' = -d_s c Q.$$

The weighted bounds also imply $\Pi(y) \to 0$. Consequently $(\Pi - 2d_s c M m)' = O(d^{-4s})$. The separation law makes d^{-4s} integrable precisely when $2s > 1$, giving the finite center in (1.9).

Finally, Section 7 moves a vertical plane from the left. The lower and lateral phase limits start the comparison. The two-layer approximation and the convergent midpoint confine possible negative minima at the upper end, uniformly for all planes on one side of b. A narrow-strip principle and positivity in the stable phases allow continuation up to b. Applying the same argument from the opposite side gives equality there, and the strong antisymmetric principle gives strict horizontal monotonicity.

The range $s > 1/2$ enters at several linked points: finite first moments for the massive drift kernel, integrable extension stress at large height, and the summability of the final center error. Each use is kept explicit in the proof.

2. Layer estimates and extension identities

Throughout the paper we write

$$\alpha = 2s \in (1, 2), \qquad a = 1 - 2s, \qquad L_1 = (-\partial_{xx})^s, \qquad L_2 = (-\Delta)^s.$$

We use $\widehat{h}(\xi) = \int_{\mathbb{R}^n} e^{-iX\cdot\xi} h(X) \, dX$, with the inverse Fourier transform normalized by $(2\pi)^{-n}$. The fractional Laplacian is normalized by

$$(-\Delta)^s h(X) = C_{n,s} \, \mathrm{PV} \int_{\mathbb{R}^n} \frac{h(X) - h(X')}{|X - X'|^{n+2s}} \, dX', \qquad C_{n,s} = \frac{4^s \Gamma(n/2 + s)}{\pi^{n/2} |\Gamma(-s)|}. \tag{2.1}$$

In particular,

$$C_{1,s} = \frac{\Gamma(1 + \alpha) \sin(\pi\alpha/2)}{\pi}.$$

The letters C and C_j denote finite constants, whose values may change. They may depend on s, c, F and the fixed solution, but not on the height or the separation of the interfaces; in auxiliary statements they may also depend on the other fixed data specified there. The speed c always retains its meaning in (1.2).

2.1. Bounded derivatives and boundary fluxes.

We first obtain the uniform bounds needed when translating the solution and removing extension cutoffs.

Lemma 2.1. *Let $1 < \alpha < 2$ and let $f \in C^2_{\mathrm{loc}}(\mathbb{R})$. Every bounded classical solution of*

$$L_2 h - c h_y + f(h) = 0 \quad \text{in } \mathbb{R}^2$$

belongs to $C^3_b(\mathbb{R}^2)$. The same conclusion holds in one dimension, with or without a constant drift.

Proof. Let $p_t^{(n)}$ be the symmetric α-stable heat kernel in dimension $n = 1$ or 2, and omit the superscript when the dimension is fixed. It is a positive probability density, with $p_t(X) = t^{-n/\alpha} p_1(t^{-1/\alpha} X)$. The density estimates and dimension-raising identity in [5, Lemmas 3.2–3.3 and Theorem 3.7, preprint version] give, in radial coordinates,

$$|\partial_r p_1^{(n)}(r)| = 2\pi r p_1^{(n+2)}(r) \le C r (1 + r)^{-n-2-\alpha}.$$

Thus $\nabla p_1 \in L^1$, and scaling gives $\|\nabla p_t\|_1 \le Ct^{-1/\alpha}$. For $\mu > 0$ define

$$\Gamma_\mu(X) = \int_0^\infty e^{-\mu t} p_t(X + cte_y)\, dt. \tag{2.2}$$

In one dimension use $p_t(x + ct)$ instead. Fourier transformation and Tonelli's theorem give, in dimension two,

$$\widehat{\Gamma}_\mu(\xi) = (\mu + |\xi|^\alpha - ic\xi_y)^{-1}, \qquad \Gamma_\mu > 0, \qquad \|\Gamma_\mu\|_1 = \mu^{-1}.$$

Moreover, differentiation in distributions and Minkowski's inequality give

$$\|\nabla\Gamma_\mu\|_1 \le C \int_0^\infty e^{-\mu t} t^{-1/\alpha}\, dt < \infty,$$

where integrability at zero uses $\alpha > 1$.

We establish the bounded inverse formula for any bounded classical solution of $T_\mu h = H$, with H bounded and $T_\mu = L_2 - c\partial_y + \mu$; the same argument applies in one dimension. Let $X_t = X + Z_t + cte_y$, where Z_t is symmetric stable, and let τ_R be the first exit from $B_R(X)$. The generator is $\mathcal{A} = -L_2 + c\partial_y$. To apply Dynkin's formula to h, choose smooth cutoffs χ_N, equal to one on $B_N(X)$ and zero outside $B_{2N}(X)$, with $0 \le \chi_N \le 1$. For $N > 2R$,

$$\sup_{B_R(X)} |\mathcal{A}(\chi_N h) - \mathcal{A}h| \le C\|h\|_\infty N^{-\alpha}.$$

Indeed, the local drift error vanishes there and the fractional error is an integral over the complement of $B_N(X)$. First mollify $\chi_N h$ and remove the mollification at fixed N; its compact support and C^2 regularity justify this passage. Then let $N \to \infty$. The displayed bound controls the stopped generator, while bounded convergence controls the value at the exit, including jumps outside the ball. Multiplying the martingale in [6, Lemma 12.2] by $e^{-\mu t}$ and integrating by parts up to $\tau_R \wedge T$ yields

$$h(X) = \mathbb{E}_X\left[e^{-\mu(\tau_R\wedge T)} h(X_{\tau_R\wedge T})\right] + \mathbb{E}_X \int_0^{\tau_R\wedge T} e^{-\mu t} H(X_t)\, dt.$$

For fixed T, every càdlàg path has bounded range on $[0, T]$, so $\tau_R \wedge T = T$ for all sufficiently large R, almost surely. Bounded convergence therefore removes the stopping. The remaining terminal term is bounded by $e^{-\mu T}\|h\|_\infty$. Letting $T \to \infty$ and using the transition density proves

$$h = \Gamma_\mu * H.$$

This representation requires no condition at spatial infinity.

It also gives comparison for bounded distributional subsolutions in either dimension. If real $h, H \in L^\infty$ satisfy $T_\mu h \le H$, convolve with a nonnegative smooth mollifier ρ_ε. Then $h_\varepsilon = \rho_\varepsilon * h$ is smooth with bounded derivatives, and $T_\mu h_\varepsilon$ is bounded. The inverse formula and positivity therefore give

$$h_\varepsilon = \Gamma_\mu * (T_\mu h_\varepsilon) \le \Gamma_\mu * (\rho_\varepsilon * H) = \rho_\varepsilon * (\Gamma_\mu * H).$$

The last convolution tends uniformly to $\Gamma_\mu * H$, which is uniformly continuous since $\Gamma_\mu \in L^1$ and H is bounded. Passing to Lebesgue points of h gives $h \le \Gamma_\mu * H$ almost everywhere, and everywhere when h is continuous.

Apply the inverse formula with $H = \mu h - f(h)$. Translation continuity in L^1 shows that convolution of $D\Gamma_\mu$ with any bounded function is bounded and uniformly continuous. In particular, $Dh = (D\Gamma_\mu) * H$ has these properties. The chain rule gives $DH = (\mu - f'(h))Dh$, which is bounded. Consequently $D^2h = (D\Gamma_\mu) * DH$ is bounded and continuous, and

$$D^2H = (\mu - f'(h))D^2h - f''(h)Dh \otimes Dh$$

is bounded as well. One more convolution gives $D^3h = (D\Gamma_\mu) * D^2H$, again bounded and uniformly continuous. All convolution differentiations first hold in distributions; their continuous representatives are the classical derivatives. Only the first two derivatives of f on the bounded range of h are used. □

The Poisson extension of a bounded trace h on $\mathbb{R}^n$ is

$$\mathcal{P}h(X,z) = P_z^{(n)} * h(X), \qquad P_z^{(n)}(X) = \frac{\Gamma(n/2+s)}{\pi^{n/2}\Gamma(s)} \frac{z^{2s}}{(|X|^2+z^2)^{(n+2s)/2}}.$$

We use $U = \mathcal{P}u$. It satisfies

$$\operatorname{div}(z^a \nabla U) = 0 \quad (z>0), \qquad -\lim_{z\downarrow 0} z^a U_z = d_s L_2 u = d_s(cu_y - F'(u)), \qquad d_s = \frac{2^{1-2s}\Gamma(1-s)}{\Gamma(s)}. \tag{2.3}$$

This is the extension of [3]; our d_s is the reciprocal of the coefficient multiplying the conormal derivative in [1, Theorem 3.1 and Remark 3.11(b)].

The next observation allows us to integrate a bottom boundary term over an unbounded vertical line. Uniform convergence of the conormal factor alone would not justify this operation.

Lemma 2.2. *Let $n \in \{1,2\}$ and $h \in C_b^2(\mathbb{R}^n)$. If $A_z = z^a \partial_z(\mathcal{P}h)$ and $A_0 = -d_s(-\Delta)^s h$, then*

$$\|A_z - A_0\|_\infty \longrightarrow 0, \qquad \|A_z\|_\infty \le C \quad (0 < z \le 1).$$

In dimension two, suppose in addition that $h = u$, $u_y > 0$, and $u(x,\pm\infty) = \pm 1$ for every x. Then, for every $R < \infty$,

$$\sup_{|x|\le R} \int_{\mathbb{R}} |U_y(x,t,z) - u_y(x,t)|\, dt \longrightarrow 0.$$

Consequently $A_z U_y \to A_0 u_y$ in $L^1(dt)$, uniformly for $|x| \le R$, both on the full line and on either fixed half-line.

Proof. Put $p_{n,s} = \Gamma(n/2+s)/(\pi^{n/2}\Gamma(s))$. The conormal derivative has the even zero-mass kernel

$$K_z^{(n)}(H) = p_{n,s}\left[\frac{\alpha}{(|H|^2+z^2)^{(n+\alpha)/2}} - \frac{(n+\alpha)z^2}{(|H|^2+z^2)^{(n+\alpha+2)/2}}\right].$$

It satisfies $|K_z^{(n)}(H)| \le C|H|^{-n-\alpha}$ and

$$K_z^{(n)}(H) \longrightarrow \alpha p_{n,s}|H|^{-n-\alpha} = d_s C_{n,s}|H|^{-n-\alpha} \qquad (H \neq 0).$$

The last identity follows directly from the constants in (2.1) and (2.3). Zero mass and evenness give

$$A_z(X) = \frac{1}{2}\int_{\mathbb{R}^n} K_z^{(n)}(H)[h(X+H) + h(X-H) - 2h(X)]\, dH.$$

The bracket is bounded uniformly by $C\min\{|H|^2, 1\}$. Therefore

$$\|A_z - A_0\|_\infty \le C\int_{\mathbb{R}^n} |K_z^{(n)}(H) - d_s C_{n,s}|H|^{-n-\alpha}| \min\{|H|^2,1\}\, dH \longrightarrow 0.$$

The majorant is integrable at zero because $\alpha < 2$ and at infinity because $\alpha > 0$. It also bounds $\|A_z\|_\infty$ uniformly, proving both assertions.

For the remaining assertions set $n = 2$ and $h = u$. Since bounded second derivatives make u_y uniformly continuous, differentiation under the Poisson integral and the approximate-identity property give $U_y = P_z^{(2)} * u_y \to u_y$ uniformly. Positivity and Tonelli's theorem, together with the vertical limits, give

$$U_y \ge 0, \qquad \int_{\mathbb{R}} u_y(x,t)\, dt = \int_{\mathbb{R}} U_y(x,t,z)\, dt = 2.$$

The continuous tails $2 - u(x,T) + u(x,-T)$ decrease to zero on every compact x interval. Dini's theorem makes this convergence uniform. Moreover, splitting the convolution into $|H| \le 1$ and $|H| > 1$ and using its positivity gives, for $T > 1$ and $0 < z \le 1$,

$$\begin{aligned}\sup_{|x|\le R}\int_{|t|>T} U_y(x,t,z)\, dt &\le \sup_{|x'|\le R+1}\int_{|t'|>T-1} u_y(x',t')\, dt' + 2\int_{|H|>1} P_z^{(2)}(H)\, dH \\ &\le \sup_{|x'|\le R+1}\int_{|t'|>T-1} u_y(x',t')\, dt' + Cz^\alpha.\end{aligned}$$

First choose T large and then z small. Together with convergence on bounded intervals this proves the L^1 assertion. Finally,

$$\int_{\mathbb{R}} |A_z U_y - A_0 u_y|\,dt \le 2\|A_z - A_0\|_\infty + \|A_0\|_\infty \int_{\mathbb{R}} |U_y - u_y|\,dt,$$

which proves both the full-line and half-line statements. □

2.2. **The one-dimensional layer and its tails.**

Lemma 2.3. *Under (1.1) there is a unique normalized increasing layer g satisfying*

$$L_1 g + F'(g) = 0, \qquad g(0) = 0, \qquad g(-\infty) = -1, \quad g(+\infty) = 1. \tag{2.4}$$

It belongs to $C_b^3(\mathbb{R})$, $q = g' > 0$, and $\int q = 2$.

Proof. Choose a smooth compactly supported cutoff equal to one near $[-1,1]$, and multiply $-F'$ by it to obtain a globally regular nonlinearity f_0. Its potential G_0, defined by $G_0' = -f_0$ and $G_0(-1) = 0$, agrees with F on $[-1,1]$. In particular, $G_0 > G_0(\pm 1) = 0$ in $(-1,1)$ and $f_0'(\pm 1) = -F''(\pm 1) < 0$. Thus [2, Theorem 2.4, preprint version] gives an increasing layer, with strictly positive derivative, unique up to translation. Its range is $(-1,1)$, so it solves the original layer equation. The classical regularity in [2, Lemma 2.1, preprint version] permits application of Lemma 2.1, which gives C_b^3. Strict increase and the end limits give a unique zero, fixing the translation, and integration of g' gives $\int g' = 2$. □

Write $\kappa_\pm = F''(\pm 1) > 0$. We need the coefficients, and not only the orders, of the algebraic tails. We obtain them from the massive one-dimensional kernel.

Lemma 2.4. *For $\kappa > 0$ let K_κ be the inverse Fourier transform of $(\kappa + |\xi|^\alpha)^{-1}$. It is positive, even, continuous and in $W^{1,1}(\mathbb{R})$, with $\int K_\kappa = \kappa^{-1}$. If $\theta = \pi\alpha/2$, then for $x > 0$,*

$$K_\kappa(x) = \frac{\sin\theta}{\pi} \int_0^\infty \frac{e^{-rx} r^\alpha}{r^{2\alpha} + 2\kappa\cos\theta\, r^\alpha + \kappa^2}\,dr. \tag{2.5}$$

As $x \to +\infty$,

$$K_\kappa(x) \sim \frac{C_{1,s}}{\kappa^2} x^{-1-\alpha}. \tag{2.6}$$

For $j = 0, 1, 2$, $|K_\kappa^{(j)}(x)| \le C_j x^{-1-\alpha-j}$ when $x \ge 1$.

Proof. The Fourier formula in the proof of Lemma 2.1 identifies K_κ with the one-dimensional, zero-drift kernel Γ_κ. Positivity, membership in $W^{1,1}$, and mass κ^{-1} therefore follow from that proof. The multiplier $(\kappa + |\xi|^\alpha)^{-1}$ is integrable because $\alpha > 1$, so its inverse is continuous and even.

For $x > 0$, write

$$K_\kappa(x) = \frac{1}{\pi} \operatorname{Re} \int_0^\infty \frac{e^{ixr}}{\kappa + r^\alpha}\,dr.$$

Rotate this integral through the first quadrant, with $z^\alpha = |z|^\alpha e^{i\alpha \arg z}$ and $0 \le \arg z \le \pi/2$. There is no pole, since $0 \le \arg z^\alpha \le \pi\alpha/2 < \pi$. On a quarter-annulus, the small arc is $O(\epsilon)$ and the large arc is $O(R^{1-\alpha})$, using $|e^{ixz}| \le 1$. Removing the arcs and taking the real part gives (2.5).

Write

$$D_\kappa(r) = r^{2\alpha} + 2\kappa\cos\theta\, r^\alpha + \kappa^2 = (r^\alpha + \kappa\cos\theta)^2 + \kappa^2 \sin^2\theta.$$

This denominator is bounded below by the positive constant $\kappa^2 \sin^2\theta$. Differentiating the Laplace integral on compact subsets of $x > 0$ and putting $r = t/x$ gives, for $j = 0, 1, 2$,

$$x^{1+\alpha+j}(-1)^j K_\kappa^{(j)}(x) = \frac{\sin\theta}{\pi} \int_0^\infty \frac{e^{-t} t^{\alpha+j}}{D_\kappa(t/x)}\,dt.$$

The denominator bound supplies an integrable majorant independent of $x \ge 1$, proving all the derivative estimates. For $j = 0$, dominated convergence gives the limit $\Gamma(1+\alpha)\sin\theta/(\pi\kappa^2) = C_{1,s}/\kappa^2$, which is (2.6). □

Proposition 2.5. *Let*

$$A_\pm = \frac{2C_{1,s}}{\alpha\kappa_\pm} = \frac{C_{1,s}}{s\kappa_\pm}.$$

As $r \to \infty$,

$$1 - g(r) \sim A_+ r^{-\alpha}, \qquad 1 + g(-r) \sim A_- r^{-\alpha}, \tag{2.7}$$

$$g'(\pm r) \sim \frac{2C_{1,s}}{\kappa_\pm} r^{-1-\alpha}. \tag{2.8}$$

Moreover, for $k = 1, 2, 3$,

$$|g^{(k)}(x)| \le C\langle x\rangle^{-\alpha-k}, \qquad \langle x\rangle = (1+x^2)^{1/2}. \tag{2.9}$$

Proof. Set $V_g = F''(g)$. Since $g \in C_b^3$, differentiation of the symmetric second-difference formula for $L_1 g$ is justified by quadratic increments near zero and bounded derivatives in the far tail. It gives $(L_1 + V_g)q = 0$. Applying the one-dimensional, zero-drift bounded inverse formula from Lemma 2.1 to $(L_1 + \lambda)q = (\lambda - V_g)q$ gives, for every $\lambda > 0$,

$$q = K_\lambda * ((\lambda - V_g)q).$$

Choose λ so small that $V_g \ge 2\lambda$ outside a fixed interval $I = [-R_0, R_0]$, with $R_0 \ge 1$. Then $S = (\lambda - V_g)_+ q$ is bounded, nonnegative, and supported in I. Positivity of K_λ gives $q \le K_\lambda * S$. For $|x| > 2R_0$, the kernel bound is uniform for $t \in I$ and yields $(K_\lambda * S)(x) \le C|x|^{-1-\alpha}\|S\|_1$. On the remaining compact interval use the boundedness of q. Thus

$$0 < q(x) \le C\langle x\rangle^{-1-\alpha}, \qquad |g(x) - \operatorname{sgn} x| \le C\langle x\rangle^{-\alpha} \quad (|x| \ge 1).$$

At the positive end put $\kappa = \kappa_+$ and $f = (\kappa - V_g)q$. The preceding bounds imply

$$f(t) = O(t^{-1-2\alpha}) \ (t \to +\infty), \qquad f(t) = O(|t|^{-1-\alpha}) \ (t \to -\infty).$$

Also

$$\int_{\mathbb{R}} f = \kappa \int q - \int F''(g)g' = 2\kappa. \tag{2.10}$$

Here $\int F''(g)g' = F'(1) - F'(-1) = 0$ by the end limits. In $q(x) = \int K_\kappa(x-t)f(t)\,dt$, first integrate over $t \le x/2$. The quantity $x^{1+\alpha}K_\kappa(x-t)$ is uniformly bounded there, converges pointwise to $C_{1,s}/\kappa^2$, and is dominated against $|f|$. On $t > x/2$ the integral is $O(x^{-1-2\alpha})\|K_\kappa\|_1$. This proves $q(x) \sim 2C_{1,s}\kappa^{-1}x^{-1-\alpha}$. Integrating gives the first formula in (2.7).

For the derivative bounds, $K_\kappa' \in L^1$ gives $q' = K_\kappa' * f$ in distributions. This convolution is bounded and continuous by L^1 translation continuity of K_κ', so the identity also holds pointwise. On $t \le x/2$ the kernel derivative estimate gives $Cx^{-2-\alpha}\|f\|_1$; on $t > x/2$ the source bound gives $Cx^{-1-2\alpha}\|K_\kappa'\|_1$. Since $\alpha > 1$,

$$|q'(x)| \le Cx^{-2-\alpha} \qquad (x \ge 2).$$

Apply the same argument to $-g(-x)$ with the potential $F(-\cdot)$. It gives the first-order asymptotics and the derivative bound at the negative end, without a symmetry assumption on F.

Finally,

$$f' = -F'''(g)q^2 + (\kappa - V_g)q'$$

is bounded and integrable, has an integrable first moment, and satisfies

$$f'(t) = O(t^{-2-2\alpha}) \ (t \to +\infty), \qquad f'(t) = O(|t|^{-2-\alpha}) \ (t \to -\infty).$$

Since $f(\pm\infty) = 0$, $\int f' = 0$. Placing the derivative on $f \in W^{1,1}$ gives $q'' = K_\kappa' * f'$ in distributions and pointwise, by the same continuity argument. Subtracting the vanishing mass term $K_\kappa'(x)\int f'$ gives, for $x \ge 2$,

$$q''(x) = \int_{\mathbb{R}} [K_\kappa'(x-t) - K_\kappa'(x)]f'(t)\,dt.$$

On $t \le x/2$ the mean value theorem bounds this integral by $Cx^{-3-\alpha}\int |tf'(t)|\,dt$. On $t > x/2$ the convolution term is bounded by $Cx^{-2-2\alpha}\|K_\kappa'\|_1$, and the subtracted term by $Cx^{-3-3\alpha}$. Both are $O(x^{-3-\alpha})$ because $\alpha > 1$. Thus $|q''(x)| \le Cx^{-3-\alpha}$. Reflection gives the negative-end bound. Boundedness of the first three derivatives on compact intervals now proves (2.9). □

3. Global geometry and orthogonal coordinates

Unless a lemma or corollary states its own hypotheses on u, we assume throughout this section that u satisfies the hypotheses of Theorem 1.1. We write $\alpha = 2s \in (1,2)$ and $a = 1-\alpha$. Under these hypotheses, Lemma 2.1 gives $u \in C_b^3(\mathbb{R}^2)$. The extension $U = P_z^{(2)} * u$ therefore has the normalization and boundary limits established in Lemma 2.2.

Lemma 3.1. *The vertical derivative has finite total dissipation:*

$$D_* := \int_{\mathbb{R}^2} u_y^2 \, dx\, dy < \infty.$$

Moreover,

$$\sup_{y\in\mathbb{R}} u_y(x,y) \longrightarrow 0 \qquad \text{as } |x| \longrightarrow \infty.$$

Proof. Lemma 2.2 gives $U_y \geq 0$ and $\int_{\mathbb{R}} U_y(x,y,z)\, dy = 2$. The scaling $P_z^{(n)}(X) = z^{-n} P_1^{(n)}(X/z)$ also gives

$$\partial_{X_i} P_z^{(n)} = z^{-n-1}(\partial_i P_1^{(n)})(X/z), \qquad \partial_z P_z^{(n)} = z^{-n-1} Q_n(X/z),$$

where $Q_n = -nP_1^{(n)} - X\cdot\nabla P_1^{(n)}$. The two profiles decay respectively as $(1+|X|)^{-n-\alpha-1}$ and $(1+|X|)^{-n-\alpha}$, and their integrals vanish. Thus all first derivative kernels have L^1 norm at most C/z. Together with the bounded trace gradient, these facts imply

$$\|U_x(\cdot,z)\|_\infty \leq C\min(1,z^{-1}), \qquad 0 \leq U_y \leq C\min(1,z^{-1}), \qquad \|U_z(\cdot,z)\|_\infty \leq C/z.$$

In particular the mixed flux

$$H(x) = \int_0^\infty \int_{\mathbb{R}} z^a U_x U_y \, dy\, dz$$

is absolutely convergent, uniformly in x, since

$$|H(x)| \leq C\int_0^1 z^{1-\alpha}\, dz + C\int_1^\infty z^{-\alpha}\, dz < \infty.$$

The extension equation yields

$$\partial_x(z^a U_x U_y) = \partial_y\left[\frac{z^a}{2}(U_x^2 - U_y^2 + U_z^2)\right] - \partial_z(z^a U_z U_y).$$

We first integrate with a compactly supported test function in x, over a finite interval in y, and over $\varepsilon < z < Z$. To remove the y boundaries, keep ε, Z fixed. The monotone convergence of u to either well is uniform on bounded horizontal intervals, by Dini's theorem, and remains so after bounded translations. For $\sigma \in \{-1,1\}$, use

$$\partial_j U(X,z) = \int_{\mathbb{R}^2} \partial_j P_z^{(2)}(Y)[u(X-Y)-\sigma]\, dY, \qquad j = x,y,z.$$

On bounded Y the integrand tends locally uniformly to zero. On its complement the L^1 tails of the derivative kernels are uniformly small for $\varepsilon \leq z \leq Z$. Hence all three derivatives vanish at the vertical boundaries, locally uniformly in x, and the corresponding stress terms disappear.

Lemma 2.2 directly gives the bottom-flux limit, locally uniformly in x:

$$\int_{\mathbb{R}} z^a U_z U_y\, dy \longrightarrow -d_s \int_{\mathbb{R}} (cu_y - F'(u))u_y\, dy.$$

The potential term integrates to $\int F'(u)u_y\, dy = F(1) - F(-1) = 0$. At the upper boundary,

$$\left|\int_{\mathbb{R}} Z^a U_z U_y\, dy\right| \leq 2Z^{1-\alpha} C/Z \leq CZ^{-\alpha}.$$

Letting $\varepsilon \downarrow 0$ and $Z \to \infty$ therefore gives

$$H'(x) = -d_s c D(x), \qquad D(x) = \int_{\mathbb{R}} u_y(x,y)^2\, dy$$

in distributions. Since $0 \le D(x) \le 2\|u_y\|_\infty$, the bounded flux has a locally Lipschitz, nonincreasing representative. Its two end limits are finite, and integration of the identity gives

$$D_* = \frac{H(-\infty) - H(+\infty)}{d_s c} \le \frac{2\|H\|_\infty}{d_s c} < \infty.$$

Finally, u_y is globally Lipschitz. If its stated horizontal limit failed, points with $|x| \to \infty$ would have $u_y \ge \eta > 0$. A fixed disk around each such point would carry a fixed positive integral of u_y^2. Selecting disjoint disks contradicts $D_* < \infty$. □

Lemma 3.2. *Let $n = 1$ or $n = 2$, $0 < \alpha < 2$, $s = \alpha/2$, and let a bounded C^2 function ψ be antisymmetric across $x_1 = \lambda$. Suppose, on $\Sigma_\lambda = \{x_1 < \lambda\}$,*

$$(-\Delta)^{\alpha/2}\psi - c\,\partial_y\psi + k\psi = 0, \qquad |k| \le K.$$

For $n = 1$ the drift term is absent. If ψ attains a negative minimum over the half-space at $X_0 \in \Sigma_\lambda$, and $d = \lambda - (X_0)_1$, then

$$(-\Delta)^{\alpha/2}\psi(X_0) \le C_{*,n} d^{-\alpha}\psi(X_0).$$

Here

$$C_{*,1} = \frac{2C_{1,s}}{\alpha}, \qquad C_{*,2} = 2C_{2,s}\int_{\eta_1<0} \frac{d\eta}{[(1-\eta_1)^2 + \eta_2^2]^{(2+\alpha)/2}} > 0.$$

Such a minimum is impossible if $k(X_0) \ge 0$; it is also impossible when $d < \delta$ and $C_{*,n}\delta^{-\alpha} > K$. If $\psi \ge 0$ in Σ_λ, then either $\psi \equiv 0$ or $\psi > 0$ there.

Proof. Reflect Y to Y^λ, and set

$$k_1 = |X_0 - Y|^{-n-\alpha}, \qquad k_2 = |X_0 - Y^\lambda|^{-n-\alpha}.$$

For $Y \in \Sigma_\lambda$, $k_1 > k_2$. Pairing the two halves of the fractional integral gives the integrand

$$[\psi(X_0) - \psi(Y)](k_1 - k_2) + 2\psi(X_0)k_2.$$

Its first term is nonpositive at a global minimum. The second kernel has integral $d^{-\alpha}/\alpha$ in one dimension. In two dimensions the substitution $Y = (\lambda + d\eta_1, (X_0)_2 + d\eta_2)$ gives the integral displayed in the statement times $d^{-\alpha}$. The local integral is legitimate because $\nabla\psi(X_0) = 0$ and the increments are quadratic; the far integral converges by boundedness. The reflected kernel is nonsingular at an interior point.

At the minimum the drift derivative vanishes. Multiplying the negative value by $k(X_0) \ge 0$ preserves the contradiction with the equation. In a strip of width δ, the same argument gives

$$[(-\Delta)^{\alpha/2} + k]\psi(X_0) \le (C_{*,n}\delta^{-\alpha} - K)\psi(X_0) < 0.$$

If instead $\psi \ge 0$ and $\psi(X_0) = 0$ at an interior point, the paired integral becomes

$$-C_{n,s}\int_{\Sigma_\lambda} \psi(Y)(k_1 - k_2)\,dY.$$

It is strictly negative unless the function vanishes on the half-space. The equation rules out strict negativity, and antisymmetry extends vanishing to the whole space. □

Lemma 3.3. *Let $-1 \le w \le 1$ be a bounded classical solution of $L_1 w + F'(w) = 0$, and let V_w be its extension. If*

$$\int_{\mathbb{R}} \left\{ \frac{1}{2}\int_0^\infty z^a[(V_w)_x^2 + (V_w)_z^2]\,dz + d_s F(w) \right\} dx < \infty,$$

then $w \equiv 1$, $w \equiv -1$, or w is a translate of $g(x)$ or $g(-x)$.

Proof. We use the following elementary strong principle: if a bounded nonnegative C^2 function h solves $L_1 h + k(x)h = 0$ and $h(x_0) = 0$, then

$$L_1 h(x_0) = -C_{1,s}\int_{\mathbb{R}} \frac{h(t)}{|x_0 - t|^{1+\alpha}}\,dt = 0$$

forces $h \equiv 0$. The integral is legitimate at this minimum by the quadratic local increment bound. This applies to $1 - w$ and $1 + w$, with bounded divided-difference coefficients. Thus contact with

a well makes w constant. Otherwise $-1 < w < 1$, and Lemma 2.1 supplies bounded derivatives. The potential integral forces the distance from $w(x)$ to $\{-1, 1\}$ to vanish as $|x| \to \infty$: failure would give disjoint fixed-length intervals on which $F(w)$ has a positive lower bound. A connected half-line cannot switch between the two well neighborhoods without meeting the intervening compact range. Thus w has a well limit at each spatial end.

Suppose first that these limits are $-1, 1$. Choose R so that w lies in the two stable well intervals for $x < -R$ and $x > R$, respectively. For all sufficiently large $T > 2R$, the comparison $w(x+T) > w(x)$ holds whenever either x or $x+T$ belongs to $[-R, R]$: the values on this compact interval stay strictly between the wells, and the other argument tends uniformly to its opposite end. If both arguments lie outside that interval, a negative difference can only have both values in the same stable well. The difference tends to zero at both spatial ends, so a negative infimum is attained. At that point its equation has a positive divided-difference coefficient, whereas its fractional Laplacian is negative, a contradiction. Thus $w(x+T) - w(x) \geq 0$ for every sufficiently large shift.

Let T_* be the infimum of the shifts above which this comparison holds for every larger shift. At $T_* > 0$, continuity gives the comparison at T_*, and the strong comparison principle makes it strict: an interior contact would make the difference identically zero, contradicting the end limits and $T_* > 0$. For shifts in a bounded neighborhood of T_*, both arguments are in the same stable well outside a fixed compact interval. On that compact interval strict positivity persists under a small decrease of the shift. Any negative minimum for such a nearby shift would therefore be in a stable tail, where the preceding minimum argument excludes it. This contradicts the definition of T_*, so $T_* = 0$. Hence w is nondecreasing. Its nonnegative derivative solves $L_1 w' + F''(w)w' = 0$; the strong maximum principle gives $w' > 0$. Lemma 2.3 identifies it with a translate of g. Reversing x treats the opposite pair of end limits.

It remains to exclude equal end limits. Assume they are both -1, and set $\psi_\lambda(x) = w(2\lambda - x) - w(x)$ for $x < \lambda$. For λ far to the left, a negative comparison has $-1 < w(2\lambda - x) < w(x)$ with $w(x)$ near -1, so both values are in the minus well interval. Moreover, $\psi_\lambda \to 0$ at $-\infty$ and vanishes at the plane. Any negative infimum is therefore attained, and Lemma 3.2 with the positive stable coefficient excludes it. The planes thus start. They cannot move indefinitely: a global maximum of w, which is attained because both limits are -1, would eventually be compared with a reflected point arbitrarily far to the right, where the value tends to -1.

Let t be the finite limiting plane. If its nonnegative comparison were not identically zero, it would be strictly positive for $x < t$. Choose a narrow width $\delta > 0$ with $C_{*,1}\delta^{-\alpha} > \|F''\|_{L^\infty([-1,1])}$. For nearby planes, a negative minimum in their δ-strip is excluded by Lemma 3.2. Far to the left, a negative comparison again puts both values in the stable minus well, independently of the nearby plane. The intervening compact interval retains strict positivity by continuity. Since any negative infimum is attained, these three regions exclude one and allow continuation beyond t, a contradiction. Consequently w is symmetric about t.

Consider

$$J_w(x) = \frac{1}{2}\int_0^\infty z^a[(V_w)_x^2 - (V_w)_z^2]\,dz - d_s F(w(x)).$$

The one-dimensional conormal assertion of Lemma 2.2 gives, uniformly in x,

$$z^a(V_w)_z \longrightarrow -d_s L_1 w = d_s F'(w).$$

Its uniform bound also gives $z^a(V_w)_z^2 \leq Cz^{\alpha-1}$ for $z \leq 1$. Also $(V_w)_x = P_z^{(1)} * w' \to w'$ uniformly: use uniform continuity of w' and $\int_{|h|>\delta} P_z^{(1)}\,dh \leq C(z/\delta)^\alpha$. Thus the bottom product tends uniformly to $d_s F'(w)w'$. At large z, both extension derivatives are $O(z^{-1})$, by the derivative-kernel L^1 bounds.

Integrate

$$\partial_x\left[\frac{z^a}{2}((V_w)_x^2 - (V_w)_z^2)\right] = -\partial_z[z^a(V_w)_z(V_w)_x]$$

between two finite x values and between $z = \varepsilon, Z$. The bottom limit cancels the derivative of $d_s F(w)$; the top product is $O(Z^{-1-\alpha})$. It follows that J_w is constant. Its absolute value is

bounded by the nonnegative, integrable slice energy density, so the constant is zero. Symmetry gives $(V_w)_x(t,z)=0$, whence

$$J_w(t)=-\frac{1}{2}\int_0^\infty z^a(V_w)_z(t,z)^2\,dz-d_sF(w(t))<0,$$

a contradiction. Equal plus limits reduce to this case under $w\mapsto -w$, $F(\tau)\mapsto F(-\tau)$. □

Lemma 3.4. *Fix $\delta\in(0,1)$. If $|x_j|\to\infty$ and $|u(x_j,y_j)|\le 1-\delta$, every sequence of translates $u_j(x,y)=u(x_j+x,y_j+y)$ has a subsequence converging in C^2_{loc} to a stationary layer, independent of y. In particular, writing $u(x,\gamma_0(x))=0$ for the unique zero height at each x, one has*

$$|u_x(x,\gamma_0(x))|\ge\eta_0>0\quad(|x|\ \textit{large}),\qquad |\gamma_0'(x)|\to\infty.$$

Proof. The global C^3_b bound gives subsequential C^2_{loc} compactness. Lemma 3.1 makes the limit independent of y. Passing to the equation uses quadratic symmetric increments near the singularity and boundedness in the integrable far tail. Denote the stationary limit by $w(x)$. The normalized kernels satisfy, for $r\neq 0$,

$$C_{2,s}\int_{\mathbb{R}}(r^2+t^2)^{-(2+\alpha)/2}\,dt=C_{1,s}|r|^{-1-\alpha}.$$

Applying this identity to symmetric second differences gives $L_2w=L_1w$, hence $L_1w+F'(w)=0$.

Put $U_j=P_z^{(2)}*u_j$. To prove finite energy, introduce

$$H_j^-(x)=\int_0^\infty\int_{-\infty}^0 z^a(U_j)_x(U_j)_y\,dy\,dz.$$

The same kernel scaling and positive vertical mass as in Lemma 3.1 give the explicit uniform bound

$$|H_j^-(x)|\le 2C\left(\int_0^1 z^{1-\alpha}\,dz+\int_1^\infty z^{-\alpha}\,dz\right)=:C_H.$$

The mixed-stress identity on the lower half-plane yields

$$\begin{aligned}&\int_{-R}^R\left\{\frac{1}{2}\int_0^\infty z^a[(U_j)_x^2+(U_j)_z^2]\,dz+d_sF(u_j(x,0))\right\}dx\\&=H_j^-(R)-H_j^-(-R)+\frac{1}{2}\int_{-R}^R\int_0^\infty z^a(U_j)_y(x,0,z)^2\,dz\,dx\\&\quad+d_sc\int_{-R}^R\int_{-\infty}^0(u_j)_y^2\,dy\,dx.\end{aligned}$$

For completeness, fix j,R,ε,Z, integrate first over $-T<y<0$, and let $T\to\infty$. Local minus-phase convergence at the lower end, together with the zero-mass derivative kernels and their uniform L^1 tails, removes its boundary stress. The top flux is $O(RZ^{-\alpha})$. Lemma 2.2 gives the lower-half-line bottom product; its potential contribution is $d_sF(u_j(x,0))$. The energy densities have majorants $Cz^{1-\alpha}$ for tangential derivatives and $Cz^{\alpha-1}$ for the vertical derivative near zero, and $Cz^{-1-\alpha}$ at infinity. These permit $\varepsilon\downarrow 0$, then $Z\to\infty$.

For fixed $z>0$, compact convergence of the traces and the L^1 tails of the differentiated Poisson kernel give $(U_j)_y(x,0,z)\to 0$. Its square is dominated by

$$z^a(U_j)_y^2\le\begin{cases}Cz^{1-\alpha}, & z\le 1,\\ Cz^{-1-\alpha}, & z\ge 1.\end{cases}$$

Thus its integral on a fixed horizontal interval tends to zero. The final term is at most d_scD_*, independently of R,j. The Poisson extension of the stationary trace is exactly V_w, since $\int_{\mathbb{R}}P_z^{(2)}(r,t)\,dt=P_z^{(1)}(r)$. Fatou's lemma now bounds the limiting stationary energy on $[-R,R]$ by $2C_H+d_scD_*$. Let $R\to\infty$ and apply Lemma 3.3. The origin value excludes the two constants.

Vertical monotonicity and the end limits give a unique zero height $\gamma_0(x)$; the implicit function theorem gives its local C^2 regularity. At zero-centered translates the normalized limit is $g(x)$ or $g(-x)$, so its derivative at the origin has magnitude $g'(0)>0$. A contradicting sequence

would have one of these limits; hence, after increasing the horizontal threshold, one may take $\eta_0 = g'(0)/2$. Finally, $\gamma_0' = -u_x/u_y$, and Lemma 3.1 gives $|\gamma_0'| \to \infty$. □

Choose once and for all levels $-1 < \theta_- < 0 < \theta_+ < 1$, sufficiently close to the wells, so that $F'' \geq 2\mu_0 > 0$ on the two outer phase intervals. Vertical monotonicity defines the unique heights $u(x, \gamma_\theta(x)) = \theta$. Write

$$\mathcal{T} = \{\theta_- \leq u \leq \theta_+\}, \qquad I_y = [-R, R] \cup \{x : |x| > R,\ \theta_- \leq u(x, y) \leq \theta_+\}.$$

Lemma 3.5. *For a sufficiently large R, the derivative u_x has a fixed nonzero sign on each of the two far transition bands and $|u_x| \geq \eta > 0$ there. Each horizontal slice of either band is one interval or is empty; its width is at most $(\theta_+ - \theta_-)/\eta$. The augmented set I_y is the union of three possibly degenerate closed intervals with globally Lipschitz endpoints, with a common constant L. It has uniformly bounded total length and at most three components. Furthermore,*

$$\operatorname{dist}(x, I_y) \leq \sqrt{1 + L^2}\, \operatorname{dist}((x, y), \mathcal{T}).$$

Proof. A sequence of far transition points with $u_x \to 0$ has, by the global C_b^3 bound, a convergent translated subsequence. Lemma 3.4 identifies its limit with a layer. The possible centered values belong to $[\theta_-, \theta_+]$, whose inverse image under g is compact. The positive minimum of g' on this set contradicts $u_x \to 0$. Each far band is the connected image of $(R, \infty) \times [\theta_-, \theta_+]$, or its left analogue, under $(x, \theta) \mapsto (x, \gamma_\theta(x))$. The nonzero continuous derivative therefore has a single sign on each band.

Here are the inverse graphs, including the heights where a slice is absent. Let $M_y = \|u_y\|_\infty$ and $\kappa = \eta/M_y$. In outward coordinates $x = \sigma(R + r)$, $\sigma = \pm 1$, $r \geq 0$, put

$$f_\theta^\sigma(r) = \gamma_\theta(\sigma(R + r)), \qquad h_\theta^\sigma = f_\theta^\sigma(0), \qquad \epsilon_\sigma = \operatorname{sign}(f_\theta^\sigma)'.$$

The sign is independent of θ, and

$$\epsilon_\sigma(f_\theta^\sigma(r) - h_\theta^\sigma) \geq \kappa r.$$

Thus the range is $[h_\theta^\sigma, \infty)$ when $\epsilon_\sigma = 1$, and $(-\infty, h_\theta^\sigma]$ when $\epsilon_\sigma = -1$. Define

$$r_\theta^\sigma(y) = \begin{cases} (f_\theta^\sigma)^{-1}(y), & \epsilon_\sigma(y - h_\theta^\sigma) > 0, \\ 0, & \epsilon_\sigma(y - h_\theta^\sigma) \leq 0. \end{cases}$$

The ordinary inverse has Lipschitz constant $L = \kappa^{-1}$. Across its clipping threshold the same bound follows from

$$r_\theta^\sigma(y_{\text{in}}) \leq L|y_{\text{in}} - h_\theta^\sigma| \leq L|y_{\text{in}} - y_{\text{out}}|.$$

For $\theta_1 < \theta_2$, one has $f_{\theta_1}^\sigma < f_{\theta_2}^\sigma$ and $h_{\theta_1}^\sigma < h_{\theta_2}^\sigma$. If $\epsilon_\sigma = 1$, the inverse order is $r_{\theta_2}^\sigma \leq r_{\theta_1}^\sigma$: below the first boundary height both are zero, between the boundary heights only the first can be positive, and above both heights ordinary inverse order applies. If $\epsilon_\sigma = -1$, the order is $r_{\theta_1}^\sigma \leq r_{\theta_2}^\sigma$: above the second boundary height both are zero, between them only the second can be positive, and below both heights ordinary inverse order applies. Set

$$(a_\sigma, b_\sigma) = \begin{cases} (r_{\theta_+}^\sigma, r_{\theta_-}^\sigma), & \epsilon_\sigma = 1, \\ (r_{\theta_-}^\sigma, r_{\theta_+}^\sigma), & \epsilon_\sigma = -1. \end{cases}$$

For $r > 0$, the equivalence

$$u(\sigma(R + r), y) \geq \theta \quad \Longleftrightarrow \quad y \geq f_\theta^\sigma(r)$$

gives precisely

$$\mathcal{T}_y \cap \{\sigma(R + r) : r > 0\} = \{\sigma(R + r) : r > 0,\ a_\sigma(y) \leq r \leq b_\sigma(y)\}.$$

Indeed, the preceding inequality means $r \leq r_\theta^\sigma$ for increasing f_θ^σ, and $r \geq r_\theta^\sigma$ for decreasing f_θ^σ; the inequalities reverse for $u \leq \theta$. This also covers the clipped cases. If $b_\sigma = 0$ the far slice is empty; if $a_\sigma = 0 < b_\sigma$ it reaches the central boundary. Each nonempty slice is thus an interval. Integrating $|u_x| \geq \eta$ along it gives its claimed width bound, also when an endpoint is clipped at $r = 0$.

Adding the possibly degenerate intervals

$$J_y^\sigma = \{\sigma(R+r) : a_\sigma(y) \le r \le b_\sigma(y)\}$$

does not change the central augmentation. Its right endpoints are $R + a_+, R + b_+$, and its left endpoints are $-R - b_-, -R - a_-$. All are L-Lipschitz. Pairing the corresponding intervals at two heights gives the motion bound

$$I_t \subset \{x : \operatorname{dist}(x, I_y) \le L|t - y|\}.$$

This remains valid when intervals meet or a far interval degenerates. The total length is at most $2R + 2(\theta_+ - \theta_-)/\eta$. Every intermediate clipped inverse lies between the two extreme ones, so $\sigma(R + r_\theta^\sigma(y)) \in I_y$. For $Q = (\xi, \tau) \in \mathcal{T}$ on a far end, choose $\theta = u(Q)$. The corresponding horizontal point is $\sigma(R + r_\theta^\sigma(\tau)) = \xi$, and consequently

$$\operatorname{dist}(x, I_y) \le |x - \xi| + L|y - \tau| \le \sqrt{1 + L^2}|(x, y) - Q|.$$

For $|\xi| \le R$ the first term alone suffices. Taking the infimum over $Q \in \mathcal{T}$ proves the claim. □

Lemma 3.6. *Let $0 < \alpha < 2$, $c \ge 0$, and let h be continuous, with $0 \le h \le B$ on $\mathbb{R}^2$. Suppose on an open set Ω that*

$$(-\Delta)^{\alpha/2} h - c h_y + k h \le 0, \qquad k \ge \mu > 0,$$

either classically or in distributions. For $X \in \Omega$, put $R_X = \operatorname{dist}(X, \Omega^c)$. Then

$$h(X) \le CB\left[(1 + R_X)^{-\alpha} + e^{-\mu R_X/(2c)}\right].$$

Here C depends only on α, μ; the exponential is omitted when $c = 0$.

Proof. Replace k by μ, which preserves the inequality because $h \ge 0$, and denote the resulting operator by T_μ. The classical inequality also holds in distributions. Mollify by a positive kernel of radius $0 < \varepsilon < R_X$. Constant coefficients give $T_\mu h_\varepsilon \le 0$ on $B_{R_X - \varepsilon}(X)$, while $0 \le h_\varepsilon \le B$.

Let Z_t be the symmetric α-stable process with generator $-(-\Delta)^{\alpha/2}$, and let τ_r be the exit time of $X + Z_t + cte_y$ from $B_r(X)$. The stopped Dynkin formula, with killing rate μ [6, Lemma 12.2], gives

$$h_\varepsilon(X) \le B\,\mathbb{E}e^{-\mu\tau_r}, \qquad r = R_X - \varepsilon.$$

The spatial-cutoff argument in the proof of Lemma 2.1 applies to the bounded smooth function h_ε for every $\alpha > 0$: its error on the stopped ball is $O(N^{-\alpha})$. Since $T_\mu h_\varepsilon \le 0$, the source term in that formula is nonpositive. Removing the finite-time terminal term, bounded by $Be^{-\mu T}$, gives the displayed inequality. If E is an independent exponential time of rate μ, $\mathbb{E}e^{-\mu\tau_r} = \mathbb{P}(\tau_r < E)$. On $cE \le r/2$, exit before E requires $\sup_{t \le E} |Z_t| \ge r/2$. The stable maximal estimate [6, Theorem 13.1]

$$\mathbb{P}\left(\sup_{t \le T} |Z_t| \ge r/2\right) \le C \min(1, Tr^{-\alpha})$$

(apply the cited strict-threshold bound at $r/4$) therefore gives, after averaging in E,

$$\mathbb{P}(\tau_r < E) \le C\mu^{-1} r^{-\alpha} + e^{-\mu r/(2c)}.$$

Let $\varepsilon \downarrow 0$. For $R_X < 1$, use the original bound; for $R_X = \infty$, apply the estimate on arbitrarily large finite balls. This proves the stated form. □

Lemma 3.7. *For the intervals I_y above,*

$$\sup_y \left(\|u_x\|_{L^1_x \cap L^2_x} + \|u_y\|_{L^1_x \cap L^2_x} + \|F(u)\|_{L^1_x}\right) < \infty.$$

All these norms have uniformly vanishing tails on $\{\operatorname{dist}(x, I_y) > N\}$ as $N \to \infty$. In addition,

$$\sup_y \|U_y(\cdot, y, z) - u_y(\cdot, y)\|_{L^1_x} \longrightarrow 0 \qquad (z \downarrow 0).$$

Furthermore,

$$\|U_x(\cdot, y, z)\|_2 + \|U_y(\cdot, y, z)\|_2 + \|U_z(\cdot, y, z)\|_2 \le C \min(1, z^{-1/2}),$$

$$\lim_{N\to\infty}\sup_{y}\int_0^Z\int_{\operatorname{dist}(x,I_y)>N} z^a|\nabla U|^2\,dx\,dz=0\quad(Z<\infty),$$

and

$$\sup_{y}\int_Z^\infty\int_{\mathbb{R}} z^a|\nabla U|^2\,dx\,dz\le CZ^{1-\alpha}\qquad(Z\ge 1).$$

Proof. On each stable phase apply Lemma 3.6 to $1-u$ or $1+u$. Their coefficients $-F'(u)/(1-u)$ and $F'(u)/(1+u)$ are positive near the corresponding well. For a point in either outer phase, its distance to the phase complement equals its distance to $\mathcal{T}$: a nearest complement point lies on $u=\theta_-$ in the minus phase and on $u=\theta_+$ in the plus phase. Differentiation and Kato's inequality give, in distributions,

$$L_2|h|-c\partial_y|h|+F''(u)|h|\le 0,\qquad h=u_x,u_y.$$

Indeed, for $b_\varepsilon(t)=\sqrt{t^2+\varepsilon^2}$, convexity under the fractional integral gives $L_2b_\varepsilon(h)\le b_\varepsilon'(h)L_2h$, while the drift obeys the chain rule. Since $0\le b_\varepsilon(h)-hb_\varepsilon'(h)\le\varepsilon$, the linearized equation implies

$$(L_2-c\partial_y+F''(u))b_\varepsilon(h)\le\|F''(u)\|_\infty\varepsilon.$$

Letting $\varepsilon\downarrow 0$ against nonnegative compactly supported tests proves the assertion; the bounded traces control the nonlocal tails. The same barrier, with bounded derivatives as exterior data, gives

$$1-u^2+|u_x|+u_y\le C(1+\operatorname{dist}((x,y),\mathcal{T}))^{-\alpha}\quad\text{outside }\mathcal{T}.$$

Inside use the global derivative bound. The distance estimate in Lemma 3.5, its component and length bounds, and $\alpha>1$ now give the asserted boundary norms and tails. For the potential use $F(u)\le C(1-u^2)^2$ near the wells.

We also record the L^1_x approximate-identity assertion needed for the horizontal bottom flux. The L^2_x bounds alone do not justify this product limit, so we prove the L^1_x convergence separately. For $h=u_y$, the preceding tails, the uniformly bounded length and number of components of I_y, and its Lipschitz motion imply

$$\sup_y\|h(\cdot-r,y-t)-h(\cdot,y)\|_1\longrightarrow 0\qquad\text{as }(r,t)\to(0,0).$$

Indeed, first discard the two tails outside a fixed large neighborhood of I_y, enlarging it by $|r|+L|t|$ for the shifted slice. The length of this neighborhood is uniformly bounded. On it the global bound on ∇h bounds the integral of the difference by $C_N(|r|+|t|)$. The difference has a uniform L^1 bound for all shifts. Integrating it against $P_z^{(2)}(r,t)$, whose mass outside any fixed disk tends to zero, proves the asserted convergence.

To transfer these conclusions to the extension, write $X=(r,t)$ and introduce the vector kernel

$$P_z(X)=\frac{\alpha}{2\pi}\frac{z^\alpha}{(z^2+r^2+t^2)^{1+\alpha/2}},\qquad \mathbf{B}_z(X)=-\frac{X}{z}P_z(X).$$

Both scale as z^{-2} times their unit-height profiles, and

$$\partial_zP_z=-\frac{2}{z}P_z-\frac{X}{z}\cdot\nabla_XP_z=\operatorname{div}_X\mathbf{B}_z.$$

Since $|\mathbf{B}_1(X)|=O(|X|^{-1-\alpha})$, $\|\mathbf{B}_z\|_1=C_B<\infty$ for $\alpha>1$. Integration by parts is justified by this integrability and bounded ∇u, and gives

$$U_x=P_z*u_x,\qquad U_y=P_z*u_y,\qquad U_z=\mathbf{B}_z*\nabla u.$$

Let $B_z(t)=(t^2+z^2)^{1/2}$. Direct integration in r gives

$$\|P_z(\cdot,t)\|_2=Cz^\alpha B_z(t)^{-3/2-\alpha},$$

$$\begin{aligned}\|\mathbf{B}_z(\cdot,t)\|_2^2&=Cz^{2\alpha-2}\int_{\mathbb{R}}\frac{r^2+t^2}{(r^2+B_z(t)^2)^{2+\alpha}}\,dr\\&\le Cz^{2\alpha-2}B_z(t)^{-1-2\alpha}.\end{aligned}$$

Consequently

$$\int_{\mathbb{R}} (\|P_z(\cdot,t)\|_2 + \|\mathbf{B}_z(\cdot,t)\|_2)\,dt \le Cz^{-1/2}.$$

The dimensionless tail powers are $|t|^{-3/2-\alpha}$ and $|t|^{-1/2-\alpha}$. For $0 < z \le Z < T$, this gives

$$\int_{|t|>T} \|P_z(\cdot,t)\|_2\,dt \le CZ^{\alpha}T^{-\alpha-1/2},$$
$$\int_{|t|>T} \|\mathbf{B}_z(\cdot,t)\|_2\,dt \le CZ^{\alpha-1}T^{1/2-\alpha}.$$

The pointwise horizontal bounds

$$P_z(r,t) \le Cz^{\alpha}|r|^{-2-\alpha}, \qquad |\mathbf{B}_z(r,t)| \le Cz^{\alpha-1}|r|^{-1-\alpha}$$

also yield, for $N > 2Z$,

$$\int_{-T}^{T} \|\mathbf{1}_{\{|r|>N/2\}}P_z(\cdot,t)\|_2\,dt \le CTZ^{\alpha}N^{-\alpha-3/2},$$
$$\int_{-T}^{T} \|\mathbf{1}_{\{|r|>N/2\}}\mathbf{B}_z(\cdot,t)\|_2\,dt \le CTZ^{\alpha-1}N^{-\alpha-1/2}.$$

Uniformity at $z = 0$ in the second estimate uses $\alpha > 1$.

For either boundary derivative h, express

$$(P_z * h)(\cdot,y) = \int_{\mathbb{R}} P_z(\cdot,t) *_x h(\cdot,y-t)\,dt.$$

Choose $T > Z$ first. The portion $|t| > T$ is small in L^2_x uniformly in y, z, by the source L^1_x bound and the vertical kernel tails. On $|t| \le T$, split the source outside and inside the $N/4$ neighborhood of I_{y-t}. The outside part is controlled by its small L^2_x tail and the full L^1 kernel norm. The Lipschitz endpoints give

$$I_{y-t} \subset \{\mathrm{dist}(x, I_y) \le L|t|\}.$$

For $N > 4LT$, the inside source lies within $N/2$ of I_y. An observation point outside distance N then requires a horizontal convolution offset larger than $N/2$; the truncated kernel bounds above and the source L^1_x norm control it. Take $N > \max(4LT, 2Z)$ after choosing T. The identical decomposition for $\mathbf{B}_z * \nabla u$, with its own tail estimates and total L^1 norm C_B, treats U_z. The resulting unweighted L^2_x tails are uniform for $0 < z \le Z$; integrating against z^a, whose integral over this interval is finite, gives bounded-height tightness.

Finally, the full L^1 kernel bounds and source L^2_x bounds give a constant bound for all derivative slice norms. The mixed $L^1_t(L^2_r)$ kernel bounds and source L^1_x bounds give the alternative $Cz^{-1/2}$. Squaring and integrating at large height gives

$$C\int_Z^{\infty} z^{1-\alpha}z^{-1}\,dz = \frac{C}{\alpha-1}Z^{1-\alpha}.$$

□

Proposition 3.8. *The zero level is an upward U: $\gamma_0(x) \to +\infty$ as $x \to \pm\infty$. For large y its inverse branches $\ell^0_1(y) < \ell^0_2(y)$ satisfy*

$$\ell^0_1 \to -\infty, \qquad \ell^0_2 \to +\infty, \qquad (\ell^0_i)' \to 0,$$

$$\|u(\cdot,y) - g(\cdot - \ell^0_1(y)) - g(\ell^0_2(y) - \cdot) + 1\|_{\infty} \to 0.$$

The lower minus phase is uniform in x. For every fixed Y, the lateral minus phase is uniform for $y \le Y$, and

$$\sup_{y\le Y} \|u(\cdot,y) + 1\|_1 < \infty.$$

Proof. The two proper monotone ends of the zero graph leave four possibilities: two one-arm orientations, an upward U, or a downward U. We distinguish them by the finite slice Hamiltonian

$$\rho(y) = \frac{1}{2}\int_{\mathbb{R}\times(0,\infty)} z^a(U_x^2 + U_z^2 - U_y^2)\,dx\,dz + d_s\int_{\mathbb{R}} F(u(x,y))\,dx.$$

Lemma 3.7 makes this expression finite. The extension stress equation and the boundary equation give

$$\rho'(y) = d_s c\int_{\mathbb{R}} u_y(x,y)^2\,dx > 0.$$

Here one may first insert a horizontal cutoff χ_R with $|\chi_R'| \le C/R$. Its cross-stress terms vanish by Cauchy–Schwarz and absolute slice-energy integrability. At height Z, the flux is bounded by

$$Z^a\|U_z(\cdot,y,Z)\|_2\|U_y(\cdot,y,Z)\|_2 \le CZ^{-\alpha}.$$

At the bottom the conormal limit is uniform. Also $U_y \to u_y$ in L^1_x, uniformly in y, by the approximate-identity assertion of Lemma 3.7. The potential derivative is integrable because F' is bounded on $[-1,1]$ and u_y has a uniform L^1_x bound. These observations justify all boundary limits in the distributional identity. On a compact height interval the moving sets I_y remain in a fixed bounded set. The uniform spatial and large-z tails and local derivative bounds give continuity of ρ by truncation to compact boxes. For the small-height truncation, the uniform L^2_x derivative bound in Lemma 3.7 gives explicitly

$$\sup_y \int_0^\varepsilon \int_{\mathbb{R}} z^a|\nabla U|^2\,dx\,dz \le C\varepsilon^{2-\alpha} \qquad (0<\varepsilon\le 1).$$

Thus ρ itself is the locally absolutely continuous representative, and the displayed derivative makes it strictly increasing.

Let G be the extension of g, and define

$$e_s = \frac{1}{2}\int_{\mathbb{R}\times(0,\infty)} z^a(G_x^2 + G_z^2)\,dx\,dz + d_s\int_{\mathbb{R}} F(g)\,dx.$$

This number is finite and positive. Indeed, the layer tails give $g' \in L^1\cap L^2$ and $F(g(x)) = O(|x|^{-2\alpha})$. In one trace dimension the vector kernel

$$B_z^{(1)}(r) = -z^{-1}(r/z)P_1^{(1)}(r/z)$$

satisfies $\partial_z P_z^{(1)} = \partial_r B_z^{(1)}$, so $G_x = P_z^{(1)} * g'$ and $G_z = B_z^{(1)} * g'$. The profile $B_1^{(1)}$ decays as $|r|^{-\alpha}$ and belongs to $L^1\cap L^2$. The kernel L^1 norms are bounded, and their L^2 norms are $O(z^{-1/2})$. Young's inequality therefore gives

$$\|G_x\|_2 + \|G_z\|_2 \le C\min(\|g'\|_2, z^{-1/2}\|g'\|_1).$$

The weighted energy is bounded at zero by $Cz^{1-\alpha}$ and at infinity by $Cz^{-\alpha}$. Finally $F(g) > 0$ on a nonempty interval, so $e_s > 0$.

We next determine ρ at either vertical endpoint. Fix $y_j \to +\infty$ or $y_j \to -\infty$. Each far zero half-graph has a half-line range with a finite boundary height. It consequently supplies either no core or one core at this endpoint. A surviving core escapes to its horizontal end, and two surviving cores separate without bound. The same description holds simultaneously for $\theta \in [\theta_-,\theta_+]$: the slope lower bound and orientation are common, and the finite boundary heights vary continuously over this compact level interval. Thus every far transition interval is eventually absent or contains the corresponding zero core, with uniformly bounded width. The sets I_{y_j} lie in the fixed central interval and fixed-width neighborhoods of the surviving cores.

At a zero core, every sequence has a C^2_{loc} subsequence by the global C^3_b bound. Lemma 3.4 identifies its limit with a layer. The zero at the origin fixes its translation, and the fixed sign of u_x fixes its orientation. All such subsequences have the same limit; hence the whole centered sequence converges. On a bounded central set, compactness and the fixed-x vertical limit similarly give convergence to a well. For $\delta \le z \le Z$, extension gradients converge locally as well. In fact,

$$|\partial_{X_i} P_z^{(2)}(X)| \le C_{\delta,Z}(1+|X|)^{-3-\alpha}, \qquad |\partial_z P_z^{(2)}(X)| \le C_{\delta,Z}(1+|X|)^{-2-\alpha}.$$

Use local trace convergence on bounded offsets and these uniform L^1 tails outside them. In the core limits U_y vanishes, and in the central well limits all gradients vanish. The uniform small-height energy estimate just proved removes $0 < z < \delta$, with error $O(\delta^{2-\alpha})$, also after centering at a core. On $\delta \le z \le Z$, local gradient convergence gives convergence of the energy on each fixed core box. Letting $\delta \downarrow 0$ justifies the bottom limit. To pass from these boxes to the whole slice, let $W = (\theta_+ - \theta_-)/\eta$. The containment of I_{y_j} proved above shows that its N-neighborhood is contained in the N-neighborhood of $[-R, R]$ together with the $(N + W)$-neighborhoods of the surviving zero cores. For fixed N these boxes are disjoint for large j. First choose Z large using the large-height energy bound, and then choose N large using bounded-height tightness and the potential slice tails in Lemma 3.7. The energy outside these boxes is arbitrarily small, uniformly in j. Inside them the preceding local convergence and small-height estimate give zero at the central well and the corresponding truncated layer energy at each core. Finally let $N, Z \to \infty$, using the finite energy of G. It follows that

$$\rho(y_j) \longrightarrow k e_s,$$

where k is the number of surviving cores.

For a one-arm graph the two endpoint values are both e_s; for a downward U they are $2e_s$ and 0. Both violate strict increase. The upward U, with endpoint values 0 and $2e_s$, is the only possibility. For $y > \max_{|x|\le R} \gamma_0(x)$, there are no central zeros, and each proper monotone end supplies exactly one zero. These inverse branches escape to opposite ends, and $(\ell_i^0)' = 1/\gamma_0'(\ell_i^0) \to 0$. Since $u(x, y) > 0$ exactly when $y > \gamma_0(x)$, the slice is positive between the two zeros and negative outside them.

For the uniform lower and lateral limits, reuse the phase bound proved in Lemma 3.7. In the minus phase it gives

$$u(X) + 1 \le C(1 + \operatorname{dist}(X, \mathcal{T}))^{-\alpha}.$$

All fixed transition levels tend upward on both ends. For each Y, choose B_Y so that $I_y \subset [-B_Y, B_Y]$ for $y \le Y$. The distance comparison then gives

$$u(x, y) + 1 \le C(1 + (|x| - B_Y)_+)^{-\alpha}, \qquad y \le Y.$$

This proves both the uniform lateral limit and its uniform L^1_x consequence. If $Y_* = \min_x \gamma_{\theta_-}(x)$, then $\mathcal{T} \subset \{y \ge Y_*\}$, and for $y < Y_*$,

$$\sup_x (u(x, y) + 1) \le C(1 + Y_* - y)^{-\alpha}.$$

This is the uniform lower phase.

Finally, the claimed uniform two-layer approximation follows by exhausting possible failure points. Within bounded distance of a zero core, use its full normalized local convergence. At bounded x away from the cores, use the locally uniform vertical plus phase. Points escaping horizontally and remaining far from both cores have distance tending to infinity from I_y; their side relative to the cores and the phase barrier force the appropriate well. Each case agrees with the stated two-layer profile, excluding a nonvanishing uniform error. □

Lemma 3.9. *Let $g \in C_b^3$ be increasing with $g' > 0$ and $g', g'', g''' \in L^1$, let $\ell_1, \ell_2 \in C^1$, and let $u \in C_b^2$. Set*

$$\Phi = g(x - \ell_1) + g(\ell_2 - x) - 1, \quad v = u - \Phi, \quad Z_1 = g'(x - \ell_1), \quad Z_2 = g'(\ell_2 - x), \quad M = \int_{\mathbb{R}} (g')^2.$$

If $\int v Z_i = 0$, then, with $O = \int Z_1 Z_2$,

$$\left(M - \int v g''(x - \ell_1)\, dx\right) \ell_1' - O\ell_2' = -\int u_y Z_1\, dx,$$

$$O\ell_1' - \left(M - \int v g''(\ell_2 - x)\, dx\right) \ell_2' = -\int u_y Z_2\, dx.$$

If $|g'(r)| \le C(1+|r|)^{-1-\alpha}$ and $d = \ell_2 - \ell_1$ tends to infinity, then $O = O(d^{-1-\alpha})$. After reversing the sign of the second row, the coefficient matrix equals $MI + O(\|v\|_\infty) + O(d^{-1-\alpha})$. Moreover, the orthogonality map

$$\Psi_i(y,p) = \int_{\mathbb{R}} (u(x,y) - \Phi_p(x)) Z_{i,p}(x)\,dx$$

is C^2 in (y, p_1, p_2). The subscript p denotes centers p_1, p_2.

Proof. The differentiation can be performed on the whole line. For $f \in W^{1,1}(\mathbb{R})$ and $b > 0$, absolute continuity and Tonelli give

$$\sup_{|h|\le b} |f(x+h)| \le |f(x)| + \int_{x-b}^{x+b} |f'(z)|\,dz =: E_{f,b}(x), \qquad \|E_{f,b}\|_1 \le \|f\|_1 + 2b\|f'\|_1.$$

On a fixed small closed height interval, the centers and their speeds are bounded. Applying this estimate to g' and g'' gives fixed integrable envelopes for the translated or reflected modes and their height derivatives. Since $v_y = u_y + \ell_1' Z_1 - \ell_2' Z_2$, all remaining factors are bounded there. Differentiation of $\int v Z_i = 0$ is justified by dominated convergence and gives the two displayed identities.

Split the overlap integral at $m = (\ell_1 + \ell_2)/2$. For $x \le m$, $Z_2 \le C d^{-1-\alpha}$, and for $x \ge m$, $Z_1 \le C d^{-1-\alpha}$. Integrating the other factor gives $O \le 2C\|g'\|_1 d^{-1-\alpha}$, including both exterior half-lines. The diagonal corrections are bounded by $\|v\|_\infty \|g''\|_1$, which proves the matrix assertion.

For the final assertion, put $d_p = p_2 - p_1$ and change variables in each integral to obtain

$$\Psi_1(y,p) = \int_{\mathbb{R}} [u(p_1 + r, y) - g(r) - g(d_p - r) + 1] g'(r)\,dr,$$
$$\Psi_2(y,p) = \int_{\mathbb{R}} [u(p_2 - r, y) - g(d_p - r) - g(r) + 1] g'(r)\,dr.$$

All parameter and height derivatives of the brackets through order two are bounded uniformly. The fixed integrable majorant $Cg'(r)$ therefore justifies differentiation and continuity of all these derivatives by dominated convergence. This proves joint C^2 regularity without differentiating g'''. □

Lemma 3.10. *For all sufficiently large y, there are unique centers $\ell_i(y)$, in a fixed small neighborhood of $\ell_i^0(y)$, such that*

$$\ell_i \in C^2, \qquad \ell_i - \ell_i^0 \to 0, \qquad \int_{\mathbb{R}} (u - \Phi) Z_i\,dx = 0.$$

They satisfy $\ell_1 \to -\infty$, $\ell_2 \to +\infty$, $\ell_i' \to 0$, and $\|u - \Phi\|_\infty \to 0$.

Proof. At a fixed height use parameters $p = (p_1, p_2)$, let $p^0 = (\ell_1^0, \ell_2^0)$, and define $\Psi_i(p; y) = \int (u - \Phi_p) Z_{i,p}$. Lemma 3.9 makes Ψ a C^2 map. If $|p - p^0| \le r$, then

$$\|u - \Phi_p\|_\infty \le \epsilon(y) + Cr, \qquad \epsilon(y) = \|u - \Phi_{p^0}\|_\infty \to 0.$$

The same parameter differentiation and overlap estimate as in Lemma 3.9 give, with $D = \operatorname{diag}(M, -M)$ and $d^0(y) = \ell_2^0(y) - \ell_1^0(y)$,

$$\sup_{|p-p^0|\le r} \|D^{-1}(D_p\Psi - D)\| \le C\big[\epsilon(y) + r + (d^0(y) - 2r)^{-1-\alpha}\big] \qquad (d^0(y) > 2r).$$

First choose a fixed small r, then a sufficiently large height, so that

$$\sup_{|p-p^0|\le r} \|D^{-1}(D_p\Psi - D)\| \le \tfrac{1}{2}.$$

The map $p \mapsto p - D^{-1}\Psi(p; y)$ is a contraction on this closed ball. Its displacement at p^0 is $O(\epsilon(y)) \le r/2$, so it maps the ball into itself. Its unique fixed point obeys $|\ell - p^0| \le C\epsilon(y) = o(1)$. The C^2 implicit function theorem and uniqueness patch the centers over all large heights.

The exact identities in Lemma 3.9 have a uniformly invertible matrix at these centers. Their right sides tend to zero. On a fixed neighborhood of a core this follows from Lemma 3.1; outside it, the L^1 tail of g' and bounded u_y give an arbitrarily small remainder. Thus $\ell_i' \to 0$. □

Corollary 3.11. *Let $u \in C_b^2(\mathbb{R}^2)$ and let $\ell_1 < \ell_2$ be C^1 centers satisfying $\int v Z_i\,dx = 0$, with the layer g and notation of Lemma 3.9. Set $g_1(r) = g(r)$, $g_2(r) = g(-r)$. At heights y_j, put $d_j = \ell_2(y_j) - \ell_1(y_j)$. Suppose that $d_j \to \infty$, $\varepsilon_j > 0$, $\varepsilon_j \to 0$, and*

$$\|v(\cdot, y_j)\|_\infty + |\ell_1'(y_j)| + |\ell_2'(y_j)| \le C\varepsilon_j, \qquad \sup_x |u_y(x, y_j)| \le C\varepsilon_j.$$

If

$$\frac{u_y(\ell_i(y_j) + r, y_j)}{\varepsilon_j} \longrightarrow \sigma_i g_i'(r) \quad \text{locally uniformly in } r,$$

then $\ell_i'(y_j)/\varepsilon_j \to -\sigma_i$.

Proof. Divide the two exact identities by ε_j. The global normalized bound and $g' \in L^1$ permit dominated convergence after centering at either core. The two right sides tend to $-\sigma_1 M$ and $+\sigma_2 M$, respectively, since $Z_2 = -g_2'$. The matrix is $\operatorname{diag}(M, -M) + o(1)$, and the normalized speeds are bounded. Its error terms therefore vanish in the limit, yielding the two assertions. □

4. The linearized layer and frozen-pair compactness

The qualitative geometry reduces the quantitative problem to the translation modes of two distant layers. We first isolate the one-layer kernel, then establish compactness after subtracting the complete pair.

Lemma 4.1 (The translation mode and its spectral gap)**.** *Let $V \in L^\infty(\mathbb{R})$ be real, and suppose that $V \ge 2\mu > 0$ outside a compact interval. Assume that a positive function*

$$q \in H^s(\mathbb{R}) \cap C_b^2(\mathbb{R}) \cap W^{2,1}(\mathbb{R})$$

satisfies $(L_1 + V)q = 0$. Then the quadratic form

$$Q_V(h) = \|(-\partial_{xx})^{s/2} h\|_2^2 + \int_{\mathbb{R}} V h^2$$

is nonnegative on $H^s(\mathbb{R})$, its nullspace is $\operatorname{span}\{q\}$, and there is $\lambda > 0$ such that

$$Q_V(h) \ge \lambda \|h\|_{H^s}^2 \qquad \text{if } \int_{\mathbb{R}} hq = 0. \tag{4.1}$$

In particular these conclusions hold for $V = F''(g)$ and $q = g'$.

Proof. We realize $L_1 + V$ on L^2 with domain H^{2s} and form domain H^s. The bounded real potential preserves self-adjointness. The weak equation for q gives $Q_V(q, \varphi) = 0$ for every $\varphi \in H^s$, in particular $Q_V(q) = 0$. For a compactly supported smooth h, write $h = q\psi$. The equation for q, tested against $q\psi^2$, gives the identity

$$Q_V(h) = \frac{C_{1,s}}{2} \iint_{\mathbb{R}^2} \frac{q(x)q(x')|\psi(x) - \psi(x')|^2}{|x - x'|^{1+\alpha}}\,dx\,dx'. \tag{4.2}$$

One may first omit $|x - x'| < \varepsilon$ in the bilinear integrals. Expansion of the squares and interchange of x, x' then prove the formula, and the second-difference bound permits $\varepsilon \downarrow 0$. Since q is positive on the support of h, the test used here is admissible by approximation. The right side is nonnegative. Density of compactly supported smooth functions in H^s extends this conclusion to the whole form domain.

If $Q_V(h) = 0$, choose compactly supported smooth $h_j \to h$ in H^s and, along a subsequence, almost everywhere. The forms $Q_V(h_j)$ tend to zero. Fatou's lemma in (4.2), first on each bounded square, shows that $h(x)/q(x)$ is almost everywhere constant. Thus the form nullspace consists precisely of the multiples of q. A distributional L^2 solution of $(L_1 + V)h = 0$ belongs to H^{2s}, because $Vh \in L^2$; hence the operator kernel has the same description.

Suppose that (4.1) fails. There are $h_j \perp q$ with $\|h_j\|_{H^s} = 1$ and $Q_V(h_j) \to 0$. After passing to a subsequence, $h_j \rightharpoonup h$ in H^s and $h_j \to h$ in L^2 on compact intervals. The Cauchy–Schwarz inequality for the nonnegative form gives

$$|Q_V(h_j, \varphi)|^2 \le Q_V(h_j) Q_V(\varphi) \longrightarrow 0 \qquad (\varphi \in H^s).$$

Passing to the weak limit shows that $(L_1+V)h=0$. The orthogonality is preserved, so $h=0$. On an interval I containing the region where $V<2\mu$, however,

$$Q_V(h_j)\geq\|(-\partial_{xx})^{s/2}h_j\|_2^2+2\mu\|h_j\|_2^2-C\|h_j\|_{L^2(I)}^2.$$

The last term tends to zero, while the sum of the first two terms is bounded below by a positive constant. This is a contradiction.

For the layer, differentiation gives $(L_1+F''(g))g'=0$. Lemma 2.3 and Proposition 2.5 give the positivity, regularity and integrability required above; $F''(g)$ is uniformly positive at both ends. The derivative bounds also place g' in H^s, for example by interpolation between L^2 and H^1. □

Lemma 4.2 (Linear growth in the transverse variable)**.** *Let $1<\alpha=2s<2$ and $c>0$. Let $V\in C_b^{1,\beta}(\mathbb{R})$ be real and satisfy $V\geq 2\mu>0$ outside a compact interval. Write $L_V=L_1+V$. Suppose that $L_V\geq 0$, that its L^2 kernel is generated by a real nonzero $q\in H^s(\mathbb{R})\cap W^{2,1}(\mathbb{R})$, and that (4.1) holds on $q^\perp$. Let $f\in C_b^{1,\beta}(\mathbb{R})$ be real. If a real classical solution of*

$$(L_2-c\partial_t+V(x))W=f(x)\qquad \text{in } \mathbb{R}^2 \tag{4.3}$$

satisfies

$$\sup_x|W(x,t)|\leq C(1+|t|), \tag{4.4}$$

then

$$W(x,t)=A(x)+t\sigma q(x),\qquad L_VA-c\sigma q=f, \tag{4.5}$$

where A is bounded and $\sigma\in\mathbb{R}$. Moreover,

$$-c\sigma\|q\|_2^2=\int_{\mathbb{R}}fq. \tag{4.6}$$

If $f=0$, then W is a constant multiple of q.

Proof. We first record the local regularity needed to take Fourier transforms in t. For a ball centered at X_0, write

$$\mathrm{Tail}_{X_0}(w)=\int_{\mathbb{R}^2}\frac{|w(X)|}{(1+|X-X_0|)^{2+\alpha}}\,dX.$$

For $(L_2-c\partial_t)w=h$ on $B_4(X_0)$, the interior estimates we use are

$$\|w\|_{C^{1,\delta}(B_1(X_0))}\leq C\big(\mathrm{Tail}_{X_0}(w)+\|w\|_{L^\infty(B_4(X_0))}+\|h\|_{L^\infty(B_4(X_0))}\big),$$
$$\|w\|_{C^{\alpha+\gamma}(B_1(X_0))}\leq C\big(\mathrm{Tail}_{X_0}(w)+\|w\|_{L^\infty(B_4(X_0))}+\|h\|_{C^{\gamma}(B_4(X_0))}\big),$$

where $0<\delta<\alpha-1$, $0<\gamma<1$, and $\alpha+\gamma$ is noninteger. The constants are independent of X_0. Corollary 2.5 of the arXiv version of [7] gives the first drift-free estimate, with exponent $1+\delta<\alpha$. Its Corollary 2.4 gives the second, initially with an additional $\|w\|_{C^\gamma}$ term. To obtain the displayed estimates, move $c\partial_t w$ to the right on nested balls and absorb this additional term as well. Interpolation gives $\|\partial_t w\|_\infty\leq\varepsilon\|w\|_{C^{1,\delta}}+C_\varepsilon\|w\|_\infty$. For the higher estimate use, instead,

$$\|w\|_{C^\gamma}+\|\partial_t w\|_{C^\gamma}\leq\varepsilon\|w\|_{C^{\alpha+\gamma}}+C_\varepsilon\|w\|_\infty.$$

Both interpolations are available because $\alpha>1$. For two nested radii $r<R$, these inequalities give $M(r)\leq\theta M(R)+C_\theta(R-r)^{-p}D$, with θ arbitrarily small, some fixed p, and D the right side of the desired estimate. Iterating over radii tending to the outer radius, with $\theta 2^p<1$, absorbs M and gives the two displayed estimates. To justify the iteration without assuming the desired regularity, mollify by a nonnegative kernel supported in B_ε, $0<\varepsilon<1/4$. The mollified equation holds on $B_{7/2}(X_0)$; its source norms on $B_3(X_0)$ and its weighted tail are bounded by the original right side. Carry out the radius iteration with $1\leq r<R\leq 3$. For fixed ε the outer smooth norm is finite, and its coefficient in the iterated inequality tends to zero. The resulting estimates are independent of ε and pass to the original solution by local convergence. A cutoff complement is smooth in the interior, with every derivative bounded by the weighted tail, as in the proofs of the cited corollaries.

The tail about (x_0, t_0) is bounded, independently of x_0, by

$$C \iint_{\mathbb{R}^2} \frac{1 + |t_0| + |\tau|}{(1 + |r|^2 + |\tau|^2)^{1+\alpha/2}} \, dr \, d\tau \le C(1 + |t_0|).$$

Here the integral of $|\tau|$ is finite since $\alpha > 1$. Apply the first interior estimate with $h = f - VW$. Its local source norm and the local norm of W have the same $C(1 + |t_0|)$ bound. Thus the first derivatives of W are bounded by $C(1 + |t_0|)$. The function $f - VW$ is now locally Lipschitz with the same bound. Choose $2 - \alpha < \gamma < 1$ and apply the higher estimate. We obtain

$$\sup_x (|\nabla W(x,t)| + |D^2 W(x,t)|) \le C(1 + |t|). \tag{4.7}$$

Fix $\vartheta \in C_c^\infty(\mathbb{R} \setminus \{0\})$, let k be its inverse Fourier transform, and put $Y = k *_t W$. The function Y may be complex. Since k is Schwartz and $\int k = 0$, equations (4.3) and (4.7) give

$$(L_2 - c\partial_t + V)Y = 0, \qquad |Y| + |\nabla Y| + |D^2 Y| \le C_\vartheta (1 + |t|). \tag{4.8}$$

The commutation with L_2 can be checked in distributions. It also follows directly from symmetric differences: use the second derivative bound at the singularity and linear growth at infinity. The latter is integrable against the kernel exactly in the range $\alpha > 1$.

We next show that Y is square integrable in x. Set $w_\delta = (|Y|^2 + \delta^2)^{1/2} - \delta$, $\delta > 0$. Convexity in the two real components of Y, applied under the positive fractional kernel, and the chain rule for the drift give

$$(L_2 - c\partial_t + \mu) w_\delta \le (\mu - V) w_\delta + C_V \delta.$$

Indeed, the difference between $|Y|^2/(|Y|^2 + \delta^2)^{1/2}$ and w_δ lies in $[0, \delta]$. Choose a nonnegative smooth compactly supported η equal to one wherever $V < 2\mu$. By (4.8), a sufficiently large C yields

$$T_\mu w_\delta \le S + C_V \delta, \qquad S(x,t) = C\eta(x)\sqrt{1 + t^2}, \quad T_\mu = L_2 - c\partial_t + \mu. \tag{4.9}$$

Let Γ_μ be the positive massive drift kernel in (2.2), with t as the drift coordinate. The stable density estimates used in Lemma 2.1 give a finite first absolute moment when $\alpha > 1$, and scaling yields

$$\iint (1 + |t|)\Gamma_\mu(x,t) \, dx \, dt \le \int_0^\infty e^{-\mu\tau}(1 + c\tau + C\tau^{1/\alpha}) \, d\tau < \infty.$$

Consequently $Z = \Gamma_\mu * S$ is finite and nonnegative, solves $T_\mu Z = S$, and satisfies

$$\|Z(\cdot, t)\|_1 + \|Z(\cdot, t)\|_\infty \le C(1 + |t|). \tag{4.10}$$

For these two bounds, integrate first in x and use, respectively, $\|\eta\|_1$ and $\|\eta\|_\infty$, together with the time marginal and its first moment. For the equation, write

$$\mathsf{P}_\tau S(X) = \int_{\mathbb{R}^2} p_\tau(H) S(X + c\tau e_t - H) \, dH.$$

Every derivative of S is bounded by $C_m(1 + |t|)$, so differentiation under the integral and the first-moment estimate give

$$|D^m \mathsf{P}_\tau S(x,t)| \le C_m(1 + |t| + c\tau + \tau^{1/\alpha}).$$

For $Z_T = \int_0^T e^{-\mu\tau} \mathsf{P}_\tau S \, d\tau$, the semigroup equation yields

$$T_\mu Z_T = S - e^{-\mu T} \mathsf{P}_T S.$$

One can first use smooth compactly supported cutoffs of S; its linear growth and $\alpha > 1$ justify their removal in the fractional integral. The displayed derivative bound permits $T \to \infty$, locally with two derivatives and in the weighted tail norm, and makes the terminal term vanish. Hence $T_\mu Z = S$.

Choose $1 < \rho < \alpha$ and let $\psi(x,t) = (1 + x^2 + t^2)^{\rho/2} + K_0$. The fractional Laplacian of its first term is bounded by $C\langle (x,t) \rangle^{\rho - \alpha}$. To see this, split the symmetric-difference integral at half the distance scale: the inner integral uses the Hessian bound $C\langle (x,t) \rangle^{\rho-2}$, and the outer integral uses

growth of order $\rho < \alpha$. The drift has size at most $C\langle(x,t)\rangle^{\rho-1}$. Thus K_0 can be chosen so that $T_\mu\psi > 0$ everywhere. For $\varepsilon > 0$ the function

$$w_\delta - Z - C_V\delta/\mu - \varepsilon\psi$$

tends to $-\infty$ at infinity. At a positive maximum its fractional Laplacian is nonnegative and its t derivative vanishes, contradicting (4.9). Let $\varepsilon \downarrow 0$ and then $\delta \downarrow 0$. We conclude that $|Y| \le Z$, so

$$\|Y(\cdot,t)\|_2^2 \le \|Z(\cdot,t)\|_\infty \|Z(\cdot,t)\|_1 \le C(1+|t|)^2. \tag{4.11}$$

The time Fourier transform $\widehat{Y}$ is therefore an L^2_x-valued tempered distribution supported in $\operatorname{supp}\vartheta$, a compact set disjoint from zero. The support assertion can also be checked by pairing with a compact spatial test function, for which time convolution multiplies the Fourier transform by ϑ, and then using density in L^2_x. At frequency ζ the equation has the operator

$$A_\zeta = (-\partial_{xx} + \zeta^2)^s + V - ic\zeta.$$

Extend Q_V sesquilinearly to complex functions. Its gap on $q^\perp$ holds for complex functions by applying the real inequality to their real and imaginary parts. The real quadratic form is

$$\operatorname{Re}\langle A_\zeta h, h\rangle = Q_V(h) + \frac{1}{2\pi}\int_{\mathbb{R}} ((\xi^2+\zeta^2)^s - |\xi|^{2s})|\widehat{h}(\xi)|^2\,d\xi. \tag{4.12}$$

This form is uniformly coercive on H^s when ζ ranges over a fixed compact set disjoint from zero. Indeed, concavity gives

$$0 \le (\xi^2+\zeta^2)^s - |\xi|^{2s} \le |\zeta|^{2s},$$

so the additional quadratic form is uniformly continuous in h in L^2 on that frequency set. Write $h = bq + h^\perp$. If a unit sequence had real form tending to zero, the gap would force $h^\perp \to 0$ in H^s. The coefficient $|b|$ would stay bounded away from zero. The second term in (4.12) is strictly positive on $q \neq 0$, continuously and uniformly on that frequency set, which is a contradiction.

The complex Lax–Milgram theorem now gives an inverse $A_\zeta^{-1} : H^{-s} \to H^s$ with uniform norm on these compact frequency sets. It depends smoothly on ζ: derivatives of $(\xi^2+\zeta^2)^s$ define bounded maps $H^s \to H^{-s}$ away from zero, and differentiation of the inverse gives the assertion. There is no domain ambiguity in applying this inverse to $\widehat{Y}$. Work on a compact frequency neighborhood disjoint from zero. Initially the equation

$$(-\partial_{xx} + \zeta^2)^s\widehat{Y} = (ic\zeta - V)\widehat{Y}$$

holds with values in H^{-2s}_x. The multiplier inverse $(-\partial_{xx}+\zeta^2)^{-s}$ is smooth as an operator $H^{-2s} \to L^2$ and as an operator $L^2 \to H^{2s}$. Applying it to the equation gives

$$\widehat{Y} = (-\partial_{xx} + \zeta^2)^{-s}(ic\zeta - V)\widehat{Y}.$$

The right side is an H^{2s}-valued distribution because $(ic\zeta - V)\widehat{Y}$ is initially L^2-valued. Thus $A_\zeta\widehat{Y} = 0$ is an H^{-s}-valued distributional identity to which the smooth inverse $A_\zeta^{-1} : H^{-s} \to H^s$ applies, giving $\widehat{Y} = 0$. Here multiplication of a vector-valued distribution by a smooth operator family is defined by transposing that family on test functions; it preserves composition of compatible operator families. All maps and all their derivatives have uniform norms on this frequency neighborhood. Hence $Y = 0$.

Every time-frequency filter supported away from zero has now vanished. Pairing W with an arbitrary compactly supported smooth function of x, its time Fourier transform is supported at zero. The structure theorem for distributions supported at a point, together with the linear growth in (4.4), makes that pairing an affine function of t. Thus $\partial_{tt}W = 0$ distributionally, and classical regularity gives

$$W(x,t) = A(x) + tB(x).$$

The functions $A = W(\cdot,0)$ and $B = W(\cdot,1) - W(\cdot,0)$ are bounded and belong to C^2_b by (4.7).

For completeness, the two-dimensional operator acts on this affine expression by

$$L_2(A + tB) = L_1A + tL_1B.$$

In the symmetric difference of tB, the additional mixed term is $-k[B(x+h)-B(x-h)]$, where (h,k) is the integration shift. It is $O(|hk|)$ near the origin and $O(|k|)$ at infinity. The former is integrable because $\alpha<2$, the latter because $\alpha>1$, and its integral vanishes by oddness in k. The remaining term reduces to L_1 with the Fourier-normalized dimensional constants. Comparing the coefficients of t in (4.3) yields

$$L_V B = 0, \qquad L_V A - cB = f. \tag{4.13}$$

Although B is bounded, it has not yet been shown to lie in the spectral space. The convex absolute-value inequality in one dimension gives, in distributions,

$$(L_1+\mu)|B| \le (\mu - V)|B| \le C\eta_1$$

for a nonnegative smooth compactly supported η_1 equal to one wherever $V<\mu$. The bounded-subsolution comparison in the proof of Lemma 2.1, in one dimension with zero drift, gives $|B| \le CK_\mu * \eta_1$. Since K_μ is integrable, $B \in L^1 \cap L^\infty$, hence $B \in L^2$. The equation places B in H^{2s}, so the kernel assumption implies $B = \sigma q$. This proves (4.5).

To extract the scalar compatibility condition, it remains to test against q without assuming $A \in L^2$. Take $\chi \in C_c^\infty(\mathbb{R})$ equal to one near zero and write $q_n = \chi(x/n)q$. Then $q_n \to q$ in $W^{2,1}$. For $h \in W^{2,1}$, absolute continuity and Tonelli give

$$\|h(\cdot+r)+h(\cdot-r)-2h\|_1 \le \min\{r^2\|h''\|_1, 4\|h\|_1\}.$$

The symmetric-difference formula, split at $|r|=1$, therefore gives

$$\|L_1 h\|_1 \le C(\|h\|_1 + \|h''\|_1).$$

It follows that $L_V q_n \to L_V q = 0$ in L^1. Symmetry of the fractional bilinear integral, justified first for smooth compact tests and then by $W^{2,1}$ approximation, yields

$$\int_{\mathbb{R}} (f + c\sigma q) q_n = \int_{\mathbb{R}} A L_V q_n \longrightarrow 0.$$

Dominated convergence on the left proves (4.6). If $f=0$, then $\sigma = 0$. The same one-dimensional massive comparison applied to A places it in L^2; (4.13) then makes A a multiple of q. □

Lemma 4.3 (Selection of a long interval)**.** *Let $d \in C^1((Y_0,\infty))$ be positive, with $d(y) \to \infty$ and $d'(y) \to 0$. Let $N \in C((Y_0,\infty))$ be nonnegative and tend to zero. If*

$$\limsup_{y\to\infty} N(y) d(y)^\alpha = \infty,$$

there are $y_j \to \infty$, $\varepsilon_j = N(y_j) > 0$ and $T_j \to \infty$ such that

$$\varepsilon_j d(y_j)^\alpha \longrightarrow \infty, \qquad \frac{T_j}{d(y_j)} \longrightarrow 0, \qquad \varepsilon_j^{-1} T_j^{-\alpha} \longrightarrow 0, \tag{4.14}$$

$$y_j - T_j > Y_0, \qquad \sup_{|t|\le T_j} N(y_j + t) \le 2\varepsilon_j. \tag{4.15}$$

Proof. Choose $z_j \to \infty$ with $N(z_j)d(z_j)^\alpha \to \infty$, and choose $k_j \to \infty$ so slowly that

$$\frac{k_j}{N(z_j)^{1/\alpha} d(z_j)} \longrightarrow 0.$$

Starting from z_j, move to a point where N is more than twice its current value whenever such a point lies within distance $k_j N^{-1/\alpha}$ of the current point. After m moves the radius is at most $k_j 2^{-m/\alpha} N(z_j)^{-1/\alpha}$. The accumulated distance, including the next search radius at any stage, is therefore bounded by

$$C_\alpha k_j N(z_j)^{-1/\alpha} = o(d(z_j)).$$

Since $d' \to 0$, we have $d(y) = o(y)$. For all sufficiently large j, every visited point and every candidate search interval lie in the fixed compact interval centered at z_j with the displayed radius, contained in (Y_0,∞). Infinitely many moves would make N unbounded on this interval, contrary to continuity. Let y_j be a terminal point and put

$$\varepsilon_j = N(y_j), \qquad T_j = k_j \varepsilon_j^{-1/\alpha}.$$

The stopping condition is exactly the supremum bound in (4.15).

All visited points lie at distance $o(d(z_j))$ from z_j. The mean-value estimate and $d' \to 0$ give $d(y_j)/d(z_j) \to 1$, while $\varepsilon_j \geq N(z_j)$. Thus $\varepsilon_j d(y_j)^\alpha \to \infty$ and

$$\frac{T_j}{d(y_j)} \leq \frac{k_j}{N(z_j)^{1/\alpha} d(y_j)} \longrightarrow 0, \qquad \varepsilon_j^{-1} T_j^{-\alpha} = k_j^{-\alpha} \longrightarrow 0.$$

We have $y_j/z_j \to 1$, so $y_j \to \infty$ and $\varepsilon_j \to 0$. Hence $T_j \to \infty$. Finally $T_j = o(d(y_j)) = o(y_j)$ gives $y_j - T_j > Y_0$ after discarding finitely many indices. □

4.1. **Static interaction and compactness of a frozen pair.** A compactness limit near either interface must retain the contribution of the other interface at the interaction scale. We therefore subtract the complete pair before taking a limit.

Lemma 4.4. *For $d \geq 2$ define*

$$\Phi_d(x) = g(x) + g(d-x) - 1, \qquad R_d = L_1 \Phi_d + F'(\Phi_d).$$

Then

$$R_d(x) = F'(g(x) + g(d-x) - 1) - F'(g(x)) - F'(g(d-x)), \qquad \|R_d\|_{C_b^1} \leq C d^{-\alpha}. \tag{4.16}$$

Locally in $C^0(\mathbb{R})$,

$$d^\alpha R_d(r) \longrightarrow A_+(\kappa_+ - F''(g(r))), \qquad d^\alpha R_d(d+r) \longrightarrow A_+(\kappa_+ - F''(g(-r))). \tag{4.17}$$

Proof. The identity follows from the layer equation. On $x \leq d/2$ put $b = g(d-x)$, $h = b - 1$ and $a_0 = g(x)$. Then $|h| \leq C d^{-\alpha}$ and $|b'| \leq C d^{-1-\alpha}$. All Taylor arguments lie in the fixed compact interval $[-3, 1]$, where the derivatives of F through order three are bounded. Taylor expansion in $F'(a_0 + h) - F'(a_0) - F'(b)$ gives the zeroth-order bound. Moreover,

$$R_d' = [F''(a_0 + h) - F''(a_0)]a_0' + [F''(a_0 + h) - F''(b)]b'.$$

This proves the derivative bound on that half-line. Reflection gives the other half. On bounded r intervals, $d^\alpha(g(d-r) - 1) \to -A_+$ uniformly, and Taylor expansion now gives $-A_+ F''(g(r)) + A_+\kappa_+$. The second limit is the reflected one. □

Here and below a moving pair is

$$\Phi(x, y) = g(x - \ell_1(y)) + g(\ell_2(y) - x) - 1, \qquad v = u - \Phi, \qquad d = \ell_2 - \ell_1. \tag{4.18}$$

For the following lemma orthogonality is not needed.

Lemma 4.5. *Let $u \in C_b^3(\mathbb{R}^2)$ solve (1.2), and let $\ell_i \in C^1((Y_0, \infty))$. Suppose $y_j \to \infty$, $d_j = d(y_j) \to \infty$, $\varepsilon_j > 0$, $\varepsilon_j \to 0$, and $T_j \to \infty$ satisfy $y_j - T_j > Y_0$ and*

$$\sup_{|t| \leq T_j} \left(\|v(\cdot, y_j + t)\|_\infty + |\ell_1'(y_j + t)| + |\ell_2'(y_j + t)| \right) \leq C\varepsilon_j, \qquad \varepsilon_j^{-1} T_j^{-\alpha} \longrightarrow 0. \tag{4.19}$$

Put $\Phi_j = \Phi(\cdot, y_j)$ and $R_j = L_1 \Phi_j + F'(\Phi_j)$, and assume $\|R_j\|_{C_b^1} \leq C\varepsilon_j$. For any sequence a_j, suppose, along a subsequence,

$$\Phi_j(a_j + \cdot) \to h_\infty \quad \text{in } C^1_{\mathrm{loc}}, \qquad \varepsilon_j^{-1} R_j(a_j + \cdot) \to r_\infty \quad \text{locally uniformly.}$$

Then a further subsequence of

$$E_j(r, t) = \frac{u(a_j + r, y_j + t) - \Phi_j(a_j + r)}{\varepsilon_j}$$

converges in $C^2_{\mathrm{loc}}(\mathbb{R}^2)$ to E, where

$$(L_2 - c\partial_t + F''(h_\infty(r)))E = -r_\infty(r), \qquad \sup_r |E(r, t)| \leq C(1 + |t|). \tag{4.20}$$

Independently of a_j, one also has

$$\sup_x (|u_y(x, y_j)| + |u_{yy}(x, y_j)|) \leq C\varepsilon_j, \qquad \|v_y(\cdot, y_j)\|_\infty \leq C\varepsilon_j. \tag{4.21}$$

Proof. Within the window, the mean value theorem and (4.19) give

$$|E_j(r,t)| \le C(1+|t|) \quad (|t| \le T_j).$$

Outside it, boundedness of u and g gives $|E_j| \le C/\varepsilon_j$. These two estimates control the nonlocal tails uniformly on every fixed ball. Indeed, integration first in the spatial displacement bounds the contribution of $|t| > T_j$ by

$$C\varepsilon_j^{-1} \int_{T_j}^{\infty} t^{-1-\alpha}\, dt \le C\varepsilon_j^{-1} T_j^{-\alpha} \longrightarrow 0.$$

Inside the window, the majorant

$$C(1+|t|)(1+r^2+t^2)^{-(2+\alpha)/2}$$

is integrable because $\alpha > 1$. The part with $S < |t| \le T_j$ is bounded by $C\int_S^{\infty}(1+t)t^{-1-\alpha}\,dt$, tending to zero as $S \to \infty$. The same integrable majorant, integrated over $r^2+t^2 > S^2$, controls escape in the horizontal direction as well. Consequently

$$\lim_{S\to\infty} \limsup_{j\to\infty} \iint_{r^2+t^2>S^2} \frac{|E_j(r,t)|}{(1+r^2+t^2)^{(2+\alpha)/2}}\, dr\, dt = 0.$$

These estimates remain valid after translating the center of any fixed observation ball, with constants depending only on that ball.

Since the frozen profile is independent of t, the normalized dimensional reduction established in Lemma 3.4 gives $L_2\Phi_j = L_1\Phi_j$. The exact equation is therefore

$$(L_2 - c\partial_t + B_j(r,t))E_j = -\varepsilon_j^{-1} R_j(a_j + r), \tag{4.22}$$

where

$$B_j(r,t) = \int_0^1 F''\big(\Phi_j(a_j+r) + \theta[u(a_j+r, y_j+t) - \Phi_j(a_j+r)]\big)\, d\theta.$$

The coefficients are uniformly bounded in $C^{1,\beta}$ on bounded sets: this uses the unnormalized C_b^3 bound on u, the corresponding bounds on g, and $F \in C^{3,\beta}_{\mathrm{loc}}$. The right side is uniformly C^1.

Use the weighted-tail interior estimates proved in Lemma 4.2, on successively smaller balls. On a larger ball the source $-B_jE_j - \varepsilon_j^{-1}R_j(a_j+r)$ is uniformly bounded, so the first estimate gives $C^{1,\delta}$ bounds on an intermediate ball, for $0 < \delta < \alpha - 1$. The source is then uniformly Lipschitz there. Choosing $2-\alpha < \gamma < 1$ with $\alpha+\gamma$ noninteger, the second estimate gives a uniform $C^{\alpha+\gamma}$ bound on the target ball, with exponent greater than two. The coefficient, residual and weighted-tail bounds make both applications uniform in j and the horizontal center a_j, even though E_j is large outside its window.

Arzelà–Ascoli and a diagonal extraction give C^2_{loc} convergence. In passing the operator to the limit, split the integral into a small ball, a bounded annulus, and its complement. Uniform second differences control the first part, local convergence controls the second, and uniform tail integrability controls the third. On compact sets $B_j \to F''(h_\infty)$, so (4.20) follows. The growth bound is inherited from the window estimate.

All the local bounds used above are independent of the choice of a_j: they use only the window bound, the uniform coefficient and residual bounds, and the weighted tail estimate. In particular this part of the argument does not require a previously chosen locally convergent background. Taking a unit ball around an arbitrary horizontal observation point at $t = 0$ bounds $\partial_t E_j$ and $\partial_{tt}E_j$ there by the same constant, which gives the first assertion of (4.21). Finally,

$$\Phi_y = -\ell_1' g'(x-\ell_1) + \ell_2' g'(\ell_2 - x),$$

and (4.19) gives the bound on v_y. □

5. Quantitative motion of the interfaces

The orthogonal coordinates of Lemma 3.10 separate the displacement of each interface from the error in its profile. We keep the notation

$$\Phi(x,y) = g(x-\ell_1(y)) + g(\ell_2(y)-x) - 1, \qquad v = u - \Phi, \qquad d = \ell_2 - \ell_1, \qquad \omega = d^{-\alpha}.$$

The qualitative approximation gives no rate for v or for the motion of the interfaces. Both rates are determined by the interaction of the two layers.

Proposition 5.1. *Let $u \in C_b^3(\mathbb{R}^2)$ solve $L_2 u - cu_y + F'(u) = 0$, with $c > 0$. Suppose $\ell_1, \ell_2 \in C^1((Y_0,\infty))$, $\ell_1 < \ell_2$, and*

$$\int_{\mathbb{R}} v Z_i \, dx = 0, \qquad d(y) \longrightarrow \infty, \qquad N(y) := \|v(\cdot,y)\|_\infty + |\ell_1'(y)| + |\ell_2'(y)| \longrightarrow 0.$$

Then $N(y) \le C\omega(y)$ for all sufficiently large y. If

$$\nu_s = \frac{2C_{1,s}}{scM}, \qquad K_* = 2\nu_s = \frac{4C_{1,s}}{scM},$$

then

$$\ell_1' = -\nu_s d^{-\alpha} + o(d^{-\alpha}), \qquad \ell_2' = \nu_s d^{-\alpha} + o(d^{-\alpha}). \tag{5.1}$$

In particular,

$$d' = K_* d^{-\alpha}(1+o(1)), \qquad d^{1+\alpha} = (1+\alpha)K_* y + o(y). \tag{5.2}$$

Proof. Lemma 4.1 supplies the spectral assumptions of Lemma 4.2 for $F''(g)$, and reflection preserves them. The required $C_b^{1,\beta}$ regularity follows from $F \in C^{3,\beta}_{\mathrm{loc}}$ and $g \in C_b^3$. The two oriented trace profiles are

$$g_1(r) = g(r), \qquad g_2(r) = g(-r).$$

Their derivatives have squared integral M and signed integrals 2 and -2, respectively.

The function N is continuous: on every compact height interval, bounded u_y and the C^1 centers make $y \mapsto v(\cdot,y)$ Lipschitz in $L^\infty(\mathbb{R})$, and the two speeds are continuous. Suppose first that N/ω is not eventually bounded. Since $d' = \ell_2' - \ell_1' \to 0$, Lemma 4.3 gives heights $y_j \to \infty$, numbers $\varepsilon_j = N(y_j)$ and windows T_j such that, with $d_j = d(y_j)$,

$$\frac{\omega(y_j)}{\varepsilon_j} \longrightarrow 0, \qquad \sup_{|t|\le T_j} N(y_j+t) \le 2\varepsilon_j, \qquad \frac{T_j}{d_j} \longrightarrow 0, \qquad \varepsilon_j^{-1} T_j^{-\alpha} \longrightarrow 0. \tag{5.3}$$

Write $\Phi_j(x) = \Phi(x,y_j)$ and $R_j = L_1\Phi_j + F'(\Phi_j)$. Lemma 4.4 yields $\|R_j\|_{C_b^1} = O(d_j^{-\alpha}) = o(\varepsilon_j)$. The separation $d_j \to \infty$ and the layer tails give

$$\Phi_j(\ell_1(y_j)+r) \to g(r), \qquad \Phi_j(\ell_2(y_j)+r) \to g(-r) \quad \text{in } C^1_{\mathrm{loc}},$$

while $R_j/\varepsilon_j \to 0$ uniformly. Freeze the entire profile Φ_j and normalize by ε_j. Applying Lemma 4.5 at both centers and extracting a common subsequence gives limits E_i satisfying

$$(L_2 - c\partial_t + F''(g_i))E_i = 0, \qquad \sup_r |E_i(r,t)| \le C(1+|t|).$$

Thus Lemma 4.2 gives $E_i = a_i g_i'$. At $t=0$ the normalized error is exactly v/ε_j on the whole line. It is uniformly bounded there. The integrability of g_i' allows passage of the orthogonality relation to the limit, and hence

$$a_i M = \int_{\mathbb{R}} E_i(r,0) g_i'(r)\, dr = 0.$$

Consequently $E_i = 0$. The local derivative convergence and the global bound for u_y/ε_j furnished by Lemma 4.5 now allow Corollary 3.11 to give $\ell_i'(y_j)/\varepsilon_j \to 0$.

The supremum norm of v must vanish at the same scale. Otherwise there are a fixed $\delta > 0$ and points x_j for which $|v(x_j,y_j)| \ge \delta\varepsilon_j$. Such points cannot remain at bounded distance from either core, by the preceding limits. After extraction, their distances from both cores tend to infinity. After another extraction, x_j lies to the left of both centers, between them, or to their right. The

frozen backgrounds then converge in C^1_{loc} to -1, 1, or -1, respectively; denote this well by τ. The normalized residuals still tend to zero. Lemma 4.5 now gives a limit E with $|E(0,0)| \ge \delta$ and

$$(L_2 - c\partial_t + \kappa_\tau)E = 0, \qquad \kappa_\tau = F''(\tau) > 0, \qquad |E(r,t)| \le C(1+|t|).$$

The power-barrier comparison in the proof of Lemma 4.2, with $\mu = \kappa_\tau$, applies to each of E and $-E$ and gives $E = 0$, contrary to $|E(0,0)| \ge \delta$. It follows that all three terms of $N(y_j)/\varepsilon_j = 1$ tend to zero, a contradiction. The bound $N \le C\omega$ is proved.

We next normalize at the interaction scale. Take any sequence $y_j \to \infty$ and put $d_j = d(y_j)$ and $\varepsilon_j = d_j^{-\alpha}$. Since $d' = o(1)$ implies $d(y) = o(y)$, we may take, for large j,

$$L_j = \min\{d_j^{\alpha/2}, (y_j/d_j)^{1/2}\} \longrightarrow \infty, \qquad T_j = d_j L_j.$$

Then $L_j d_j^{-\alpha} \to 0$, $T_j = o(y_j)$, and $y_j - T_j \to \infty$. Until a first exit of $d(y_j + t)$ from $[d_j/2, 2d_j]$, the rate bound already proved gives $|d'(y_j + t)| \le C d_j^{-\alpha}$. Integrating forward or backward from $t = 0$ bounds its relative variation by $CL_j d_j^{-\alpha} = o(1)$ throughout the window. At a first exit that variation would be at least $1/2$, a contradiction. Hence $d(y_j + t)/d_j \to 1$ uniformly on the window, $N \le C\varepsilon_j$ there, and

$$\varepsilon_j^{-1} T_j^{-\alpha} = L_j^{-\alpha} \longrightarrow 0.$$

Set $A_+ = C_{1,s}/(s\kappa_+)$. The core residual in Lemma 4.4 and the compactness in Lemma 4.5 give

$$(L_2 - c\partial_t + F''(g_i))E_i = A_+(F''(g_i) - \kappa_+), \qquad |E_i(r,t)| \le C(1+|t|).$$

By Lemma 4.2, $E_i = A_i + t\sigma_i g_i'$, where

$$\begin{aligned} -c\sigma_i M &= A_+ \int_{\mathbb{R}} (F''(g_i) - \kappa_+) g_i'\, dr \\ &= -A_+\kappa_+ \int_{\mathbb{R}} g_i'\, dr. \end{aligned}$$

Here

$$\int_{\mathbb{R}} F''(g_i) g_i'\, dr = F'(g_i(+\infty)) - F'(g_i(-\infty)) = 0.$$

The normalized vertical derivatives converge to $\sigma_i g_i'$ locally and remain bounded globally. Thus Corollary 3.11 gives

$$d_j^\alpha \ell_i'(y_j) \longrightarrow -\sigma_i = -\frac{A_+\kappa_+}{cM} \int_{\mathbb{R}} g_i'\, dr.$$

Since $A_+\kappa_+ = C_{1,s}/s$, the two signed masses give (5.1). These limits are the same along every convergent subsequence of the bounded normalized speeds, so they hold on the entire upper end. Subtracting the two identities and integrating $(d^{1+\alpha})' = (1+\alpha)K_*(1+o(1))$ proves (5.2). □

Corollary 5.2. *Under the hypotheses of Proposition 5.1, the functions ℓ_i are C^2 at large heights, and*

$$\sup_x \left(|u_y| + |u_{yy}| + \sum_{|\nu| \le 2} |\partial^\nu v| \right) + |\ell_1''| + |\ell_2''| \le C\omega. \tag{5.4}$$

Here ν is a physical derivative multi-index in (x,y). Moreover, for $0 < \gamma \le 1$,

$$[\ell_i']_{C^\gamma([y-1,y+1])} \le C_\gamma \omega(y).$$

Proof. For any $y_j \to \infty$, take $\varepsilon_j = d_j^{-\alpha}$ and the windows $T_j = d_j L_j$ constructed in the interaction-scale part of the preceding proof. They give $N \le C\varepsilon_j$ throughout the window, $\varepsilon_j^{-1} T_j^{-\alpha} \to 0$, and $y_j - T_j \to \infty$. The static residual bound supplies $\|R_j\|_{C^1_b} \le C\varepsilon_j$. The uniform local C^2 estimates in the proof of Lemma 4.5 therefore apply at every horizontal center, independently of a convergent background. They bound all first and second derivatives of the normalized frozen error by a common constant on a fixed ball at time zero. At that height, horizontal derivatives of the frozen error are those of v/ε_j, while its t, rt and tt derivatives are u_y/ε_j, u_{xy}/ε_j and u_{yy}/ε_j.

Since the sequence y_j was arbitrary, these uniform bounds hold at all sufficiently large heights. Consequently

$$\|v_x\|_\infty + \|v_{xx}\|_\infty + \|u_y\|_\infty + \|u_{xy}\|_\infty + \|u_{yy}\|_\infty \le C\omega. \tag{5.5}$$

The formula for Φ_y and the speed bound also give $\|v_y\|_\infty \le C\omega$.

To recover the regularity of the centers themselves, view the two orthogonality integrals as a map $\Psi(y, \xi_1, \xi_2)$ from Lemma 3.9. That lemma gives its C^2 regularity and the matrix $D_\xi\Psi = \operatorname{diag}(M, -M) + O(\omega) + O(d^{-1-\alpha})$ at the centers. It is uniformly invertible for large y, so the implicit-function theorem gives a local C^2 branch. The given C^1 center curve agrees with that branch by local uniqueness, so $\ell_i \in C^2$ at every sufficiently large height.

The second differentiation is legitimate in the L^∞–L^1 pairing: $g' \in W^{2,1}$ makes $y \mapsto Z_i(\cdot, y)$ a C^2 map into L^1, while $u, g \in C_b^3$ and the C^2 centers make $y \mapsto v(\cdot, y)$ locally C^2 into L^∞. Consequently

$$0 = \int_{\mathbb{R}} v_{yy} Z_i + 2\int_{\mathbb{R}} v_y (Z_i)_y + \int_{\mathbb{R}} v (Z_i)_{yy}.$$

The derivatives appearing here are

$$\begin{aligned} v_{yy} &= u_{yy} + \ell_1'' Z_1 - \ell_2'' Z_2 - (\ell_1')^2 g''(x-\ell_1) - (\ell_2')^2 g''(\ell_2 - x), \\ (Z_1)_{yy} &= -\ell_1'' g''(x-\ell_1) + (\ell_1')^2 g'''(x-\ell_1), \\ (Z_2)_{yy} &= \ell_2'' g''(\ell_2 - x) + (\ell_2')^2 g'''(\ell_2 - x). \end{aligned}$$

The coefficient of (ℓ_1'', ℓ_2'') is again $D_\xi\Psi$. The terms $\int u_{yy} Z_i$ are $O(\omega)$ by (5.5). The remaining integrals contain either two speeds, or $v_y(Z_i)_y$, or $v(\ell_i')^2 g'''$; their bounds are $O(\omega^2)$ or $O(\omega^3)$, using $g', g'', g''' \in L^1$. Inversion gives $|\ell_i''| \le C\omega$. Finally,

$$\Phi_{xy} = -\ell_1' g''(x-\ell_1) - \ell_2' g''(\ell_2 - x), \qquad \Phi_{yy} = -\ell_1'' Z_1 + \ell_2'' Z_2 + (\ell_1')^2 g''(x-\ell_1) + (\ell_2')^2 g''(\ell_2 - x).$$

These expressions and (5.5) prove (5.4). The bound for ℓ_i'' is uniform on $[y-1, y+1]$, since d is comparable there to $d(y)$. For t_1, t_2 in this interval,

$$|\ell_i'(t_1) - \ell_i'(t_2)| \le C\omega(y)|t_1 - t_2| \le 2^{1-\gamma} C\omega(y)|t_1 - t_2|^\gamma,$$

which proves the stated Hölder bounds. □

6. Weighted approximation and center convergence

The leading velocities in (5.1) cancel in m'. The remaining drift is controlled by the spatially integrated error. For this purpose, the slowly decaying exterior part of the two-layer profile must first be corrected.

Fix a sufficiently large height $Y > 1$. Composing both centers with the same smooth height map, constant below Y and equal to the identity above $Y + 1$, extends them to C^2 functions on $\mathbb{R}$. Choose Y so that $d \ge d_0 > 2$ throughout the range of this map. The extension preserves every estimate at large heights and is constant for $y \le 1$. Orthogonality will only be used where the centers have not been changed. Proposition 3.8 supplies

$$\sup_{y \le Y_1} \|u(\cdot, y) + 1\|_1 < \infty \qquad \text{for each fixed } Y_1. \tag{6.1}$$

Proposition 6.1. *Let $u \in C_b^3(\mathbb{R}^2)$, $|u| < 1$, solve $L_2 u - c u_y + F'(u) = 0$. Suppose $\ell_i \in C^2(\mathbb{R})$ are constant for $y \le 1$, $d \ge d_0 > 2$, and, at all sufficiently large heights,*

$$\|v\|_\infty + |\ell_1'| + |\ell_2'| + |\ell_1''| + |\ell_2''| \le C\omega, \qquad d' = Kd^{-\alpha}(1 + o(1)), \quad K > 0.$$

Assume also (6.1). Choose a smooth function ζ, equal to zero on $(-\infty, 0]$ and to one on $[1, \infty)$, and set

$$\begin{aligned} \mu_* &= 1 - \frac{\kappa_+}{\kappa_-}, \\ \chi(x, y) &= \mu_* \big[\zeta(\ell_1 - x)(1 - g(\ell_2 - x)) + \zeta(x - \ell_2)(1 - g(x - \ell_1))\big], \qquad \widetilde{v} = v - \chi. \end{aligned} \tag{6.2}$$

Then, for large y,

$$\|\widetilde{v}\|_{H_x^s} + \|\widetilde{v}\|_\infty + \|\widetilde{v}_x\|_2 + \|\widetilde{v}_y\|_2 \le C\omega, \tag{6.3}$$

and

$$\|v\|_\infty + \|v_x\|_2 + \|v_y\|_2 \le C\omega, \qquad \|v\|_2 \le Cd^{1/2}\omega. \tag{6.4}$$

Proof. The separation law gives

$$d(y)^{1+\alpha} \asymp 1 + y, \qquad \omega(y) \asymp h_0(y) := (1 + y_+)^{-\alpha/(1+\alpha)}$$

at large heights, and d is eventually increasing. Since $|\ell_i'| \le Cd'$ there, integration gives $|\ell_i(y)| \le C(1 + d(y))$. These bounds also control the part of the history before the observation height; no bound for m is needed at this stage. More explicitly, after increasing a fixed threshold Y_0, for $y \ge 2Y_0$ we have

$$\sup_{|t|\le y/2} \omega(y+t) \le C\omega(y), \qquad \omega(y+t) \le C\omega(y) \quad (t \ge 0), \qquad |\ell_i(t)| \le C(1 + t_+^{1/(1+\alpha)}).$$

The last bound holds on the whole extended real line. The first two center derivatives are bounded there, because the centers are constant below a fixed height and obey the assumed bounds above another fixed height.

First quantify the correction in (6.2). For $\beta_d(r) = \zeta(r)(1 - g(d + r))$, the layer estimates give

$$\|\beta_d\|_\infty \le C\omega, \qquad \|\beta_d\|_1 \le Cd^{1-\alpha}, \qquad \|\beta_d\|_2 \le Cd^{1/2-\alpha}. \tag{6.5}$$

Each $\partial_r^j \partial_d^k \beta_d$, with $1 \le j + k \le 2$, has $L^1 \cap L^\infty$ norm at most $C\omega$. Consequently

$$\|2\beta_d - \beta_d(\cdot + h) - \beta_d(\cdot - h)\|_1 \le C \min\{h^2\omega, |h|\omega, d^{1-\alpha}\},$$
$$\|2\beta_d - \beta_d(\cdot + h) - \beta_d(\cdot - h)\|_\infty \le C \min\{h^2\omega, \omega\}.$$

For the first bound, split the fractional integral at 1 and d. Its three contributions are bounded by

$$C\omega \int_0^1 h^{1-\alpha}\, dh, \qquad C\omega \int_1^d h^{-\alpha}\, dh, \qquad Cd^{1-\alpha} \int_d^\infty h^{-1-\alpha}\, dh.$$

They are $O(\omega)$ because $1 < \alpha < 2$. The supremum estimate follows from the second difference bound above, so $\|L_1\beta_d\|_{L^1\cap L^\infty} \le C\omega$. The L^2 bound and the $L^1 \cap L^\infty$ bound for β_d' put β_d in $H^1(\mathbb{R})$. Fractional integration by parts then gives

$$[\beta_d]_{\dot H^s}^2 \le C\|\beta_d\|_\infty \|L_1\beta_d\|_1 \le C\omega^2.$$

Using the two translated copies of β_d in χ and the chain rule yields

$$\begin{aligned} \|\chi\|_\infty + [\chi]_{\dot H^s} + \|\chi_x\|_{L^1\cap L^\infty} + \|\chi_y\|_{L^1\cap L^\infty} &\le C\omega, \\ \|\chi\|_2 &\le Cd^{1/2}\omega. \end{aligned} \tag{6.6}$$

In fact $\|\chi_y\|_{L^1\cap L^\infty} \le C\omega^2$ at large heights.

We next bound the equation satisfied by $\widetilde\Phi = \Phi + \chi$. The dependence of a layer on its center produces a nonlocal error in addition to the static interaction. For $\varphi(r) = g(r)$ or $g(-r)$ and $\delta_\ell(t) = \ell(y + t) - \ell(y)$, direct integration in the vertical variable gives

$$(L_2 - L_1)\varphi(x - \ell(y)) = C_{2,s} \lim_{\epsilon\downarrow 0} \int_{|t|>\epsilon} \int_{\mathbb{R}} \frac{\varphi(\xi) - \varphi(\xi - \delta_\ell(t))}{((x - \ell(y) - \xi)^2 + t^2)^{1+s}}\, d\xi\, dt. \tag{6.7}$$

To check the sign, subtract the operator with the center frozen at $\ell(y)$ before putting $\xi = x' - \ell(y)$ in the horizontal integration. The remaining numerator is the frozen value minus the value at height $y + t$, exactly as displayed. The frozen operator is L_1 by the dimensional reduction in Lemma 3.4. The symmetric limit in t is essential: its paired integrand, rather than each first difference separately, is integrable near $t = 0$. The horizontal integral of the kernel is $C|t|^{-1-\alpha}$. When $|t| < 1$, pair t and $-t$. The acceleration bound gives $\delta_\ell(t) + \delta_\ell(-t) = O(\omega t^2)$, while the sum of their squares is $O(\omega^2 t^2)$. Taylor expansion, using $\varphi', \varphi'' \in L^1 \cap L^\infty$, bounds the paired numerator in either norm by $C\omega t^2$. This part is therefore bounded by $C\omega \int_0^1 t^{1-\alpha} dt$. This also verifies that the displayed symmetric limit agrees with the radial principal value defining L_2: reflection $t \mapsto -t$ preserves both truncations, cancels the first-order term, and leaves an absolutely integrable paired expression. Dominated convergence then removes either truncation

with the same limit. For $t > 1$, monotonicity of d gives $|\delta_\ell(t)| \le Ct\omega(y)$. The same estimate holds for $-y/2 < t < -1$, since $d(y+t) \asymp d(y)$. For $t < -y/2$, the bound $|\ell(y+t)| + |\ell(y)| \le Cd(y)$ gives $|\delta_\ell(t)| \le Cd(y)$. Since

$$\|\varphi - \varphi(\cdot - \delta)\|_1 = 2|\delta|, \qquad \|\varphi - \varphi(\cdot - \delta)\|_\infty \le C|\delta|,$$

these regions contribute at most

$$C\omega \int_1^\infty t^{-\alpha}\, dt + Cd(y)y^{-\alpha} \le C\omega.$$

For the last inequality, $d(y)y^{-\alpha}/\omega(y) = O(y^{1-\alpha})$. At bounded or negative observation heights, the global parameter bounds give the paired small-time bound Ct^2. For every fixed Y_1, the global growth bound also gives, uniformly for $y \le Y_1$ and $|t| \ge 1$,

$$|\delta_\ell(t)| \le C_{Y_1}(1 + |t|^{1/(1+\alpha)}).$$

Its product with $|t|^{-1-\alpha}$ is integrable. This proves the required uniform $L^1 \cap L^\infty$ estimate on every lower height half-line, including observation heights tending to $-\infty$.

For a correction slice $B_y(x) = \beta_{d(y)}(\ell_1(y) - x)$, the corresponding identity in absolute horizontal coordinates is

$$(L_2 - L_1)B(x,y) = C_{2,s} \lim_{\epsilon \downarrow 0} \int_{|t|>\epsilon} \int_{\mathbb{R}} \frac{B_y(\xi) - B_{y+t}(\xi)}{((x-\xi)^2 + t^2)^{1+s}}\, d\xi\, dt, \qquad B(x,y) = B_y(x).$$

The parameter derivatives of β_d of orders one and two have $L^1 \cap L^\infty$ norm $O(\omega)$, while $|d'| + |d''| + |\ell_1'| + |\ell_1''| \le C\omega$ at large heights. The chain rule therefore gives

$$\|\partial_y B(\cdot, y)\|_{L^1 \cap L^\infty} + \|\partial_{yy} B(\cdot, y)\|_{L^1 \cap L^\infty} \le C\omega(y)^2.$$

Pairing t and $-t$ thus bounds the small-time numerator by $C\omega(y)^2 t^2$. In the future and in the past up to $y/2$, the first difference is bounded by $C|t|\omega(y)^2$, using the upper bounds for the weight along these intervals. In the remote past the norm is uniformly bounded by (6.5), since $\alpha > 1$ and $d \ge d_0$. The resulting motion error is at most

$$C\omega^2 \int_0^1 t^{1-\alpha}\, dt + C\omega^2 \int_1^\infty t^{-\alpha}\, dt + Cy^{-\alpha} \le C\omega.$$

The symmetric-limit justification is the same as for (6.7). Applying the same estimates to the right correction gives

$$\|(L_2 - L_1)\chi\|_{L^1 \cap L^\infty} \le C\omega. \tag{6.8}$$

At bounded and negative heights, bounded parameter derivatives give a uniform paired small-time estimate. Both correction slices have uniform $L^1 \cap L^\infty$ bounds at every height, so their large-time differences are bounded uniformly as well. This proves the uniform estimate on the entire lower height half-line.

It remains to use the coefficient μ_* in the static error. Between the centers let $r = x - \ell_1$, $\delta_1 = 1 - g(r)$ and $\delta_2 = 1 - g(d - r)$. Both deficits lie in $[0,1]$, and $\chi = 0$. The identity

$$\begin{aligned} &F'(1 - \delta_1 - \delta_2) - F'(1 - \delta_1) - F'(1 - \delta_2) + F'(1) \\ &\quad = \int_0^{\delta_1} \int_0^{\delta_2} F'''(1 - \xi - \eta)\, d\eta\, d\xi \end{aligned}$$

bounds the static interaction by $C\delta_1\delta_2 \le C(1+r)^{-\alpha}(1+d-r)^{-\alpha}$. Splitting $[0,d]$ at its midpoint gives its supremum bound $Cd^{-\alpha}$ and its integral bound

$$Cd^{-\alpha} \int_0^\infty (1+r)^{-\alpha}\, dr \le C\omega.$$

On the left exterior, write $x = \ell_1 - r$, $\delta_1 = 1 + g(-r)$ and $\delta_2 = 1 - g(d+r)$. Outside the fixed cutoff strip,

$$\widetilde{\Phi} = -1 + \delta_1 - \frac{\kappa_+}{\kappa_-}\delta_2.$$

The coefficient of δ_2 in the static interaction vanishes at $\delta_1 = 0$, because $-(\kappa_+/\kappa_-)\kappa_- + \kappa_+ = 0$. Taylor expansion gives

$$\left|F'\left(-1+\delta_1-\frac{\kappa_+}{\kappa_-}\delta_2\right) - F'(-1+\delta_1) - F'(1-\delta_2)\right| \le C(\delta_1\delta_2 + \delta_2^2).$$

The resulting exterior integral is bounded by

$$C\int_0^\infty [(1+r)^{-\alpha}(d+r)^{-\alpha} + (d+r)^{-2\alpha}]\,dr \le C\omega.$$

Reflection gives the same bound on the right exterior, with no symmetry condition on F. On each cutoff strip, the uncorrected static residual is $O(\omega)$ by (4.16), and $|F'(\Phi+\chi)-F'(\Phi)| \le C|\chi| \le C\omega$. The strips have fixed width, so they contribute $O(\omega)$ in both norms. We assemble the residual as

$$\widetilde{R} = [L_1\Phi + F'(\Phi+\chi)] + L_1\chi + (L_2-L_1)\Phi + (L_2-L_1)\chi - c(\Phi_y+\chi_y).$$

The preceding static bounds, the motion estimates (6.7)–(6.8), and $\|\Phi_y\|_{L^1\cap L^\infty} + \|\chi_y\|_{L^1\cap L^\infty} \le C\omega$ bound every term by $C\omega$ at large heights. Their uniform bounds on lower height half-lines, together with $\omega \asymp h_0$ on the upper end, give

$$\widetilde{R} := L_2\widetilde{\Phi} - c\widetilde{\Phi}_y + F'(\widetilde{\Phi}), \qquad \|\widetilde{R}(\cdot,y)\|_{L^1\cap L^\infty} \le Ch_0(y) \tag{6.9}$$

for every height y.

The exact equation for $\widetilde{v}$ is

$$L_2\widetilde{v} - c\widetilde{v}_y + \widetilde{a}\widetilde{v} = -\widetilde{R}, \qquad \widetilde{a} = \int_0^1 F''(\widetilde{\Phi}+t\widetilde{v})\,dt.$$

For large y, this coefficient is at least $2\mu_0 > 0$ outside two fixed-width core intervals. On those intervals $|\widetilde{v}| \le C\omega$. At heights bounded above, the layer tails and (6.5) give

$$\|\widetilde{\Phi}(\cdot,y) + 1\|_1 \le C(1+d(y)) :$$

the interval between the centers has length d, and the two exterior tails are integrable since $\alpha > 1$. The right side is uniformly bounded on each lower height half-line, because the extended centers are constant below 1. Together with (6.1), this gives uniform $L^1\cap L^\infty$ bounds for $\widetilde{v}$ there. Kato's inequality thus gives

$$T_{\mu_0}|\widetilde{v}| \le S, \qquad S \ge 0, \qquad \|S(\cdot,y)\|_{L^1\cap L^\infty} \le Ch_0(y), \tag{6.10}$$

where $T_\mu = L_2 - c\partial_y + \mu$. Here one may take $S = |\widetilde{R}| + (\mu_0 - \widetilde{a})_+|\widetilde{v}|$. Indeed, convex approximation of the absolute value in the singular integral gives $L_2|\widetilde{v}| \le \operatorname{sgn}(\widetilde{v})L_2\widetilde{v}$ in distributions, while the first-order drift obeys the usual chain rule. The two core intervals have uniformly bounded total length, which gives the stated slice norm of S on the upper end; the lower slice bounds give it on the rest of the line. Both $|\widetilde{v}|$ and S are bounded. The bounded-subsolution comparison proved in Lemma 2.1 therefore applies to (6.10) and gives $|\widetilde{v}| \le \Gamma_{\mu_0} * S$ everywhere.

We record how the time weight passes through this kernel. The stable density estimates and the dimension-raising identity used in the proof of Lemma 2.1 give

$$p_\tau(X) \le \frac{C\tau}{(\tau^{1/\alpha}+|X|)^{2+\alpha}}, \qquad |\nabla p_\tau(X)| \le \frac{C\tau}{(\tau^{1/\alpha}+|X|)^{3+\alpha}};$$

where stable scaling has been used. Put $k_j(t) = \|\nabla^j\Gamma_\mu(\cdot,t)\|_1$, $j = 0,1$; the same lemma gives $k_j \in L^1(\mathbb{R})$. To control their tails, integrate the displayed estimates in x:

$$\int_{\mathbb{R}} |\nabla^j p_\tau(x,b)|\,dx \le \frac{C\tau}{(\tau^{1/\alpha}+|b|)^{1+\alpha+j}} \le C\tau^{-(1+j)/\alpha}.$$

Splitting the semigroup time at $|t|/(2c)$ yields, for $|t| \ge 1$,

$$k_0(t) \le C(1+|t|)^{-1-\alpha} + Ce^{-\mu|t|/(4c)}, \qquad k_1(t) \le C(1+|t|)^{-2-\alpha} + Ce^{-\mu|t|/(4c)}.$$

On the first part, $|t+c\tau| \ge |t|/2$ and the integral is at most $C|t|^{-1-\alpha-j}\int_0^\infty e^{-\mu\tau}\tau\,d\tau$. On the second, use the bound $C\tau^{-(1+j)/\alpha}$ and the discounted exponential; here $\tau \ge 1/(2c)$, so there is no small-time singularity. For large positive y, split the weighted convolution into $|t| \le y/2$,

$t < -y/2$ and $t > y/2$. On the first two regions $h_0(y-t) \le Ch_0(y)$. On the last, $h_0 \le 1$ and the tail mass is $O(y^{-\alpha}) \le Ch_0(y)$. Bounded and negative heights follow from $k_j \in L^1$. Thus

$$k_j * h_0 \le Ch_0, \qquad j = 0, 1, \tag{6.11}$$

also for L^1_x- and L^∞_x-valued data, by Minkowski's inequality.

Equations (6.10) and (6.11) give $\|\widetilde{v}\|_{L^1\cap L^\infty} \le Ch_0$. Now write

$$T_\mu \widetilde{v} = (\mu - \widetilde{a})\widetilde{v} - \widetilde{R} =: f.$$

The source f has the same weighted slice bound. Bounded uniqueness for T_μ gives $\widetilde{v} = \Gamma_\mu * f$; differentiating the kernel and applying (6.11) gives $\|\nabla\widetilde{v}\|_{L^1\cap L^\infty} \le Ch_0$. Interpolation yields the L^2 bounds for $\widetilde{v}$ and its first physical derivatives, and interpolation between L^2_x and H^1_x yields (6.3). Adding (6.6) proves (6.4). □

Proposition 6.2. *In addition to the hypotheses of Proposition 6.1, assume $u_y > 0$, that both horizontal ends of each slice are -1, and that $\int v Z_i\,dx = 0$ for sufficiently large y. Suppose*

$$\sup_y\big(\|u_x\|_{L^1\cap L^2} + \|u_y\|_{L^1\cap L^2} + \|F(u)\|_1\big) < \infty$$

and

$$\|U_x(\cdot,y,z)\|_2 + \|U_y(\cdot,y,z)\|_2 + \|U_z(\cdot,y,z)\|_2 \le C\min\{1, z^{-1/2}\}.$$

Then $m(y) \to b$ for some $b \in \mathbb{R}$ as $y \to \infty$.

Proof. Let $Q(y) = \int_{\mathbb{R}} u_x u_y\,dx$. The profile part of this product can be evaluated exactly:

$$\int \Phi_x\Phi_y = -M(\ell_1' + \ell_2') + (\ell_1' + \ell_2')\int Z_1 Z_2 = -2Mm' + O(\omega^2).$$

Lemma 3.9 gives the overlap bound $O(d^{-1-\alpha})$ and justifies differentiating orthogonality:

$$\int v_y Z_i = -\int v(Z_i)_y = O(\omega^2), \qquad \|(Z_i)_y\|_1 \le C|\ell_i'|.$$

Since $\Phi_x = Z_1 - Z_2$, this gives $\int \Phi_x v_y = O(\omega^2)$. Moreover, $\|\Phi_y\|_2 \le C\omega$, and (6.4) gives

$$\left|\int v_x(\Phi_y + v_y)\,dx\right| \le \|v_x\|_2(\|\Phi_y\|_2 + \|v_y\|_2) \le C\omega^2.$$

Consequently

$$Q = -2Mm' + O(\omega^2). \tag{6.12}$$

For the extension (2.3), the mixed flux

$$\Pi(y) = \int_0^\infty \int_{\mathbb{R}} z^a U_x U_y\,dx\,dz$$

is absolutely convergent, since $\int_0^\infty z^{1-\alpha}\min\{1, z^{-1}\}dz < \infty$. The extension equation gives

$$\partial_y(z^a U_x U_y) + \partial_x\left[\tfrac{1}{2}z^a(U_x^2 - U_y^2 - U_z^2)\right] + \partial_z(z^a U_x U_z) = 0.$$

Integrate first with a horizontal cutoff and on $\epsilon < z < Z$, against a compactly supported test function of y. The horizontal error tends to zero by the uniform slice energy on this compact height interval and the factor $O(R^{-1})$ from the cutoff. At the top, $Z^a\|U_x\|_2\|U_z\|_2 \le CZ^{-\alpha}$; the tail of Π is $O(Z^{1-\alpha})$.

For the bottom passage, u_x is integrable on $\mathbb{R}\times I$ for every bounded height interval I. The Poisson approximate identity therefore gives $U_x \to u_x$ in this local-in-height L^1 space. One may verify this by truncating to a slightly larger height interval and controlling the omitted vertical offsets by the global slice bound and the Poisson tail mass. Put $A_z = z^a U_z$ and $A_0 = -d_s(cu_y - F'(u))$. Lemma 2.2 gives bounded A_z and $\|A_z - A_0\|_\infty \to 0$. In particular,

$$\|A_z U_x - A_0 u_x\|_{L^1(\mathbb{R}\times I)} \le \|A_z\|_\infty\|U_x - u_x\|_{L^1(\mathbb{R}\times I)} + \|A_z - A_0\|_\infty\|u_x\|_{L^1(\mathbb{R}\times I)} \longrightarrow 0.$$

This proves the required bottom product limit. Finally, $\int F'(u)u_x\,dx = \int \partial_x F(u)\,dx = 0$, since both horizontal ends are -1. Thus

$$\Pi' = -d_s \int u_x(cu_y - F'(u))\,dx = -d_s cQ. \tag{6.13}$$

The calculation first holds distributionally; the local regularity and the integrable bounds identify its locally absolutely continuous representative with the displayed flux at every height. Indeed, for fixed $z > 0$, the Poisson kernels are continuous under vertical translation in $L^1(dt; L^2_x)$, so the uniform boundary slice L^1_x bounds make $U_x(\cdot, y, z)$ and $U_y(\cdot, y, z)$ continuous in y with values in L^2_x. The integrable bound $Cz^{1-\alpha}\min(1, z^{-1})$ then makes Π continuous.

The weighted approximation also makes this flux vanish at the upper end. Indeed, $\|\Phi_y\|_2 + \|v_y\|_2 \le C\omega$, so $\|u_y(\cdot, y)\|_2 \to 0$. For fixed $z > 0$, convolution with the positive Poisson kernel, whose time marginal has mass one, implies $\|U_y(\cdot, y, z)\|_2 \to 0$. The global slice bound permits dominated convergence in the time variable. The integrable majorant $Cz^{1-\alpha}\min\{1, z^{-1}\}$ then gives

$$\int_0^\infty z^a \|U_y(\cdot, y, z)\|_2^2\,dz \longrightarrow 0.$$

The corresponding U_x energy is uniformly bounded. Cauchy–Schwarz therefore yields $\Pi(y) \to 0$.

Combining (6.12) and (6.13) gives

$$(\Pi - 2d_s cMm)' = O(\omega^2).$$

Since $d' \asymp d^{-\alpha}$,

$$\int^\infty \omega(y)^2\,dy \asymp \int^\infty D^{-\alpha}\,dD < \infty.$$

Hence $\Pi - 2d_s cMm$ has a finite limit, and $\Pi \to 0$ implies the claimed finite limit of m. □

7. Reflection and completion of the main theorems

The conclusions needed for the final reflection argument now follow from the preceding estimates. Proposition 5.1 and Corollary 5.2 give the upper-end error and derivative bounds. Use the extension of the centers fixed at the start of Section 6. Proposition 3.8 gives the lower slice integrability, horizontal minus ends, uniform lower minus phase, and uniform lateral minus phase on every lower height half-plane, while Lemma 3.7 gives all uniform boundary slice bounds and extension-energy bounds. These are the hypotheses of Proposition 6.1 and then Proposition 6.2. In particular the orthogonal centers satisfy

$$\ell_1(y) \to -\infty, \qquad \ell_2(y) \to +\infty, \qquad m(y) = \frac{\ell_1(y) + \ell_2(y)}{2} \longrightarrow b \in \mathbb{R},$$

and u converges uniformly to the associated two-layer profile. For a vertical reflection plane, write

$$x^\lambda = 2\lambda - x, \qquad \Sigma_\lambda = \{x < \lambda\}, \qquad w_\lambda(x, y) = u(x^\lambda, y) - u(x, y).$$

Reflection invariance of L_2, and its commutation with ∂_y, give

$$L_2 w_\lambda - c\partial_y w_\lambda + k_\lambda w_\lambda = 0.$$

Here

$$k_\lambda(x, y) = \begin{cases} \dfrac{F'(u(x^\lambda, y)) - F'(u(x, y))}{u(x^\lambda, y) - u(x, y)}, & u(x^\lambda, y) \ne u(x, y), \\ F''(u(x, y)), & u(x^\lambda, y) = u(x, y). \end{cases}$$

The mean value theorem supplies the continuous extension at coincident values and the global bound

$$|k_\lambda| \le K := \max_{[-1,1]} |F''|.$$

When both values lie in one stable well neighborhood, k_λ is uniformly positive.

Lemma 7.1. *Fix $\varepsilon, \delta > 0$ and stable neighborhoods*

$$N_- = [-1, -1+\eta], \qquad N_+ = [1-\eta, 1], \qquad F'' \geq \mu > 0 \quad \text{on } N_- \cup N_+.$$

There is Y such that, for every $y \geq Y$, $\lambda \leq b - \varepsilon$, and $x \leq \lambda - \delta$, a negative value of $w_\lambda(x, y)$ forces the two compared values to belong to the same one of N_- and N_+.

Proof. If the assertion fails, there are $y_j \to \infty$, $\lambda_j \leq b - \varepsilon$, and $x_j \leq \lambda_j - \delta$ with $w_{\lambda_j}(x_j, y_j) < 0$, but with the two values in no common neighborhood N_- or N_+. Suppress the index and set

$$x' = 2\lambda - x, \qquad r = x - \ell_1, \qquad h = \lambda - \ell_1, \qquad A = m - \lambda.$$

Then

$$h + A = d/2, \qquad h - r \geq \delta, \qquad A \geq \varepsilon/2$$

eventually. Uniform profile convergence allows a subsequence on which each point either remains a bounded distance from one core or escapes both cores and converges to the corresponding well. The function g is strictly increasing and $-1 < g < 1$, so all finite translated-layer comparisons below are strict.

If x remains near a core, it is the left core, because $x < \lambda \leq m - \varepsilon/2$. Take $r \to r_0$. When $h \to h_0 \in \mathbb{R}$, both comparison points are at the left core and

$$w_\lambda \longrightarrow g(2h_0 - r_0) - g(r_0) > 0, \qquad h_0 - r_0 \geq \delta.$$

When $h \to +\infty$ and $A \to A_0 < \infty$, the reflected point is at the right core, and

$$w_\lambda \longrightarrow g(r_0 + 2A_0) - g(r_0) > 0, \qquad A_0 \geq \varepsilon/2.$$

When $h, A \to +\infty$, it is in the plus plateau and the limit is $1 - g(r_0) > 0$. All three cases contradict negativity.

We may now suppose that x escapes both cores. If x' stays near the left core, put $r' = x' - \ell_1 \to r_0'$. Since $r \leq r' - 2\delta$, escape forces $r \to -\infty$. If x' instead stays near the right core, write $t = \ell_2 - x' \to t_0$; then $r = t - 2A \leq t - \varepsilon$, again forcing $r \to -\infty$. Thus the original value tends to -1, whereas the reflected value tends to $g(r_0')$ or $g(t_0)$, each strictly greater than -1. This also contradicts negativity.

The remaining case has both points escaping both cores. Equal limiting phases put both values in the same prescribed stable neighborhood. Different phases with a negative difference would require the original point to be in the plus plateau and the reflected point in the minus phase. But then

$$x - \ell_1 \to +\infty, \qquad \ell_2 - x \to +\infty, \qquad x' \geq x, \qquad \ell_2 - x' = x - \ell_1 + 2A \to +\infty,$$

so the reflected point is also in the plus plateau. This contradiction exhausts the possibilities. No bound on λ from below has been used. □

Lemma 7.2. *All sufficiently far left planes satisfy $w_\lambda \geq 0$ in Σ_λ. If $\lambda_* < b$ and $w_{\lambda_*} \geq 0$ is not identically zero, the same inequality holds for every plane sufficiently close to λ_*. Every fixed plane $\lambda > b$ has a negative reflected difference somewhere.*

Proof. For each fixed $\lambda < b$, first observe that

$$\liminf_{\substack{|(x,y)| \to \infty \\ x < \lambda}} w_\lambda(x, y) \geq 0.$$

Indeed, if y is bounded above and $x \to -\infty$,

$$w_\lambda(x, y) \geq -1 - u(x, y) \to 0$$

by the uniform lateral phase. If $y \to -\infty$, both values tend uniformly to -1. For $y \to +\infty$, outside a fixed thin strip at the plane, Lemma 7.1 can be applied with arbitrarily small stable neighborhoods of width η. A negative difference there lies in $[-\eta, 0)$. Inside the strip, fixed λ keeps both horizontal coordinates in a bounded interval, where the vertical plus limit is uniform by monotonicity and Dini's theorem. This proves the lower limit. A negative infimum, if present, is therefore attained at an interior point of Σ_λ, since $w_\lambda = 0$ on its boundary.

Choose a strip width $\delta > 0$ with $C_{*,2}\delta^{-\alpha} > K$, as in Lemma 3.2. Lemma 7.1, with $\varepsilon = 1$, gives one threshold Y for every $\lambda \le b - 1$. For λ sufficiently negative, the lateral limit puts the original value in N_- for all $x < \lambda, y \le Y$. If the difference is negative, $-1 < u < 1$ puts the reflected value in N_- as well. For $y > Y$, outside the strip both values belong to the same stable neighborhood. Thus a negative global minimum either has a positive coefficient or lies in the narrow strip. Lemma 3.2 excludes both alternatives and starts the planes.

For continuation, the strong alternative gives $w_{\lambda_*} > 0$ on the open old half-plane. Fix

$$0 < \eta_0 < \min\{\delta/2, (b - \lambda_*)/2\}, \qquad |\lambda - \lambda_*| \le \eta_0.$$

This interval remains uniformly below b. Lemma 7.1 confines possible negative minima outside their own strips from above; the uniform lower minus phase confines them from below. For y bounded above and x sufficiently negative, negativity again places both values in N_-. All remaining negative minima outside their own strips are consequently confined to a fixed bounded set. Explicitly, there are L, Y_-, Y_+, independent of these nearby planes, such that a negative minimum outside its own strip with $k_\lambda < 0$ must have

$$L \le x, \qquad Y_- \le y \le Y_+.$$

The alternatives outside this set have $k_\lambda \ge \mu$ and are already excluded by Lemma 3.2. Such a point satisfies

$$x \le \lambda - \delta \le \lambda_* + \eta_0 - \delta < \lambda_* - \delta/2.$$

The confinement set can therefore be enlarged to a fixed compact rectangle contained strictly in Σ_{λ_*}. The old difference has a positive minimum on this rectangle. Continuity in (x, y, λ) preserves positivity there for sufficiently close planes. Shrinking η_0 at this last step preserves all earlier confinement estimates. A new negative minimum must then be at a stable-coefficient point or within the narrow strip, both impossible.

Finally let $\lambda > b$ be fixed and evaluate at $x = \ell_1(y) < \lambda$ for large y. Uniform profile convergence and $m(y) \to b$ give

$$w_\lambda(\ell_1(y), y) \longrightarrow g(2(b - \lambda)) - g(0) < 0.$$

□

Proof of Theorem 1.1. First note that any plane of exact symmetry must equal b. Indeed, if $w_\lambda \equiv 0$, reflection exchanges the two zeros on each sufficiently high slice, so

$$\lambda = \frac{\ell_1^0(y) + \ell_2^0(y)}{2} = m(y) + o(1) \longrightarrow b.$$

Here we used $\ell_i - \ell_i^0 \to 0$.

Define

$$\Lambda = \{\lambda : w_\mu \ge 0 \text{ in } \Sigma_\mu \text{ for every } \mu \le \lambda\}, \qquad \bar\lambda = \sup \Lambda.$$

Lemma 7.2 makes Λ nonempty and gives $\bar\lambda \le b$. For each $\lambda < \bar\lambda$, some element of Λ lies above λ, so $w_\lambda \ge 0$. Continuity gives $w_{\bar\lambda} \ge 0$, and hence $\bar\lambda \in \Lambda$. If $\bar\lambda < b$, the strong alternative in Lemma 3.2 leaves two possibilities. Strict positivity allows continuation through an interval beyond $\bar\lambda$; together with the inequalities for every earlier plane, this places a larger plane in Λ, a contradiction. Identical vanishing would give $\bar\lambda = b$ by the preceding observation, also a contradiction. It follows that $\bar\lambda = b$ and $w_b \ge 0$.

Apply the same left-plane argument to $\widehat{u}(x, y) = u(-x, y)$. Its orthogonal centers and midpoint are

$$\widehat{\ell}_1 = -\ell_2, \qquad \widehat{\ell}_2 = -\ell_1, \qquad \widehat{d} = d, \qquad \widehat{m} = -m \longrightarrow -b.$$

The two profile terms and the two orthogonality conditions are exchanged, and the equation and all bounds are preserved with the same potential F. The resulting inequality is $u(2b - X, y) \ge u(X, y)$ for $X > b$, which is opposite to $w_b \ge 0$ after reflecting X back to the left half-plane. Thus $u(2b - x, y) = u(x, y)$ everywhere.

For every $\lambda < b$, the difference is strictly positive: the strong alternative and the observation about symmetry planes exclude vanishing. Taking the plane midway between $x_1 < x_2 < b$ yields

$u(x_2, y) > u(x_1, y)$, so $u_x \geq 0$ on $x < b$. The bounded C^2 derivative u_x is now antisymmetric about b and solves

$$L_2 u_x - c\partial_y u_x + F''(u)u_x = 0.$$

Differentiation is justified by the global C^3_b bound. The strict increase just proved rules out $u_x \equiv 0$. The strong alternative in Lemma 3.2 implies $u_x > 0$ for $x < b$. Reflection gives $u_x < 0$ for $x > b$. □

Proof of Theorem 1.2. Proposition 3.8 supplies the two zero branches, and Lemma 3.10 supplies their C^2 orthogonal centers, their vanishing differences from the zero branches, and their horizontal limits. Proposition 5.1 gives the profile bound, the signed velocities, and the separation law, with $\alpha = 2s$ and $K_* = 4C_{1,s}/(scM)$. Corollary 5.2 gives the acceleration bound. Proposition 6.2 gives $m(y) \to b$, and the preceding proof identifies b with the reflection axis. □

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